Producer statement cover sheet for Pool Fencing / Barrier

One cover sheet must be provided for each balustrade system

Auckland Council
Te Kaunihera o Tāmaki Makaurau

PART 1: TO BE COMPLETED BY THE DESIGN PROFESSIONAL / MANUFACTURER SUPPLYING THE GLASS SYSTEM

Cover sheet issued on behalf of:	Royal Glass
In respect of (system type):	Mini Post SP14 Pool Fence
Supplied to:	Auckland Council
Client / designers name:	
Description of work:	
Address of property:	

PART 2: CONSTRUCTION DETAILS *(tick boxes that apply)*

Deck / balcony / stairs construction:	☐ Timber	☐ Precast	☐ Concrete	☐ Steel	☐ *Please specify*

Maximum fall height:	< 1 (m)	**Note:** If barrier is not protecting a fall of more than 1.0m, barrier required to be designed for wind load only – no additional safety features required

PART 3: GLASS DETAILS *(tick boxes that apply)*

Type:	☐ Toughened	☐ Toughened laminated	☐ Toughened laminated with stiff interlayer

Thickness:	(mm)	Width of pane:	1.8 (m)
Height above floor:	(m)	Height above fixing:	(m)

Note: If barrier extends more than 1.2m above fixing point, test reports must be provided

Supplier:	

PART 4: BARRIER DETAILS

System name:	Mini Post SP14 Pool Fence
Drawings (ref page number)	
Engineers name:	
Hardware (type):	

Drawings must show the type of hardware being used to support the glass (i.e. channel, discs, etc.)

PART 5: SAFETY FEATURES *(tick all boxes that apply; not required if fall height <1.0m)*	N/A ☐
	☐ Continuous interlinking rail (show how fixed to building and each pane of glass)
	☐ Continuous interlinking top cap (show how fixed to building and each pane of glass)
	☐ Clamps, brackets or point fixings (show how fixed to building and each pane of glass)
	☐ Framed with structural sealant min. 2 edges (show how fixed to building and each pane of glass)
	☐ Glass with stiff interlayer
	☐ Two or three glass edges supported by continuous clamps
	☐ Structural post to fix interlinking rail to deck
	☐ Other (specify)
Engineers name:	Lautrec Technology Group
Drawings (ref page number)	
Comments:	

PART 6: COVER SHEET COMPLETED BY	
Name:	Roxy Huang
Signature:	
Date:	
Contact number:	0800 769 254
Email:	info@royalglass.co.nz

FOR RESIDENTIAL OCCUPANCY A, A OTHER & C3 OF TABLE 3.3 AS/NZS 1170.1

FOR PROTECTING A FALL OF MORE THAN 1M AND LESS THAN 5M FROM THE GROUND

NOTES:
1) This proprietary balustrade system complies with New Zealand Building Code Clause B1 Structure, B1/AS1 Amendment 15, B1/VM1, F2 Hazardous Building Materials and F4 Safety From Falling Third Addition, subject to:
- all products meeting the required performance specification
- site installation carried out in accordance with intent of this drawing and to be installed by a certified glass balustrade practitioner

For residential & commercial occupancy A, A Other, C3 of Table 3.3 AS/NZS 1170.1

2) Design of support structure is the responsibility of others

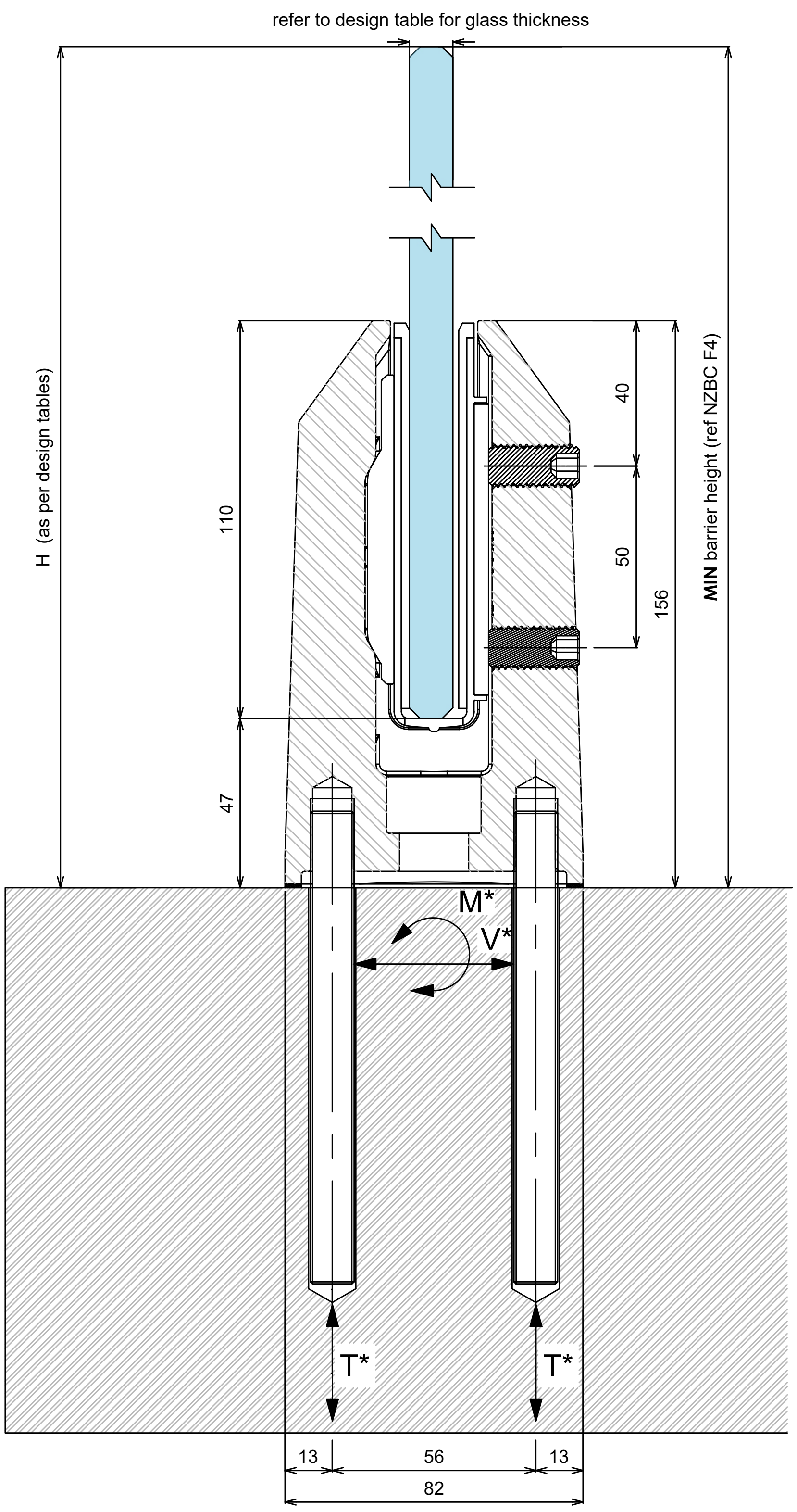
3) A handrail of 32-50mm diameter is required for stairs and ramps exceeding 1:20 slope. Refer NZBC D1/AS1

4) Safety glass Options according to 22.4.3 of NZS 4223.3:2016 are:

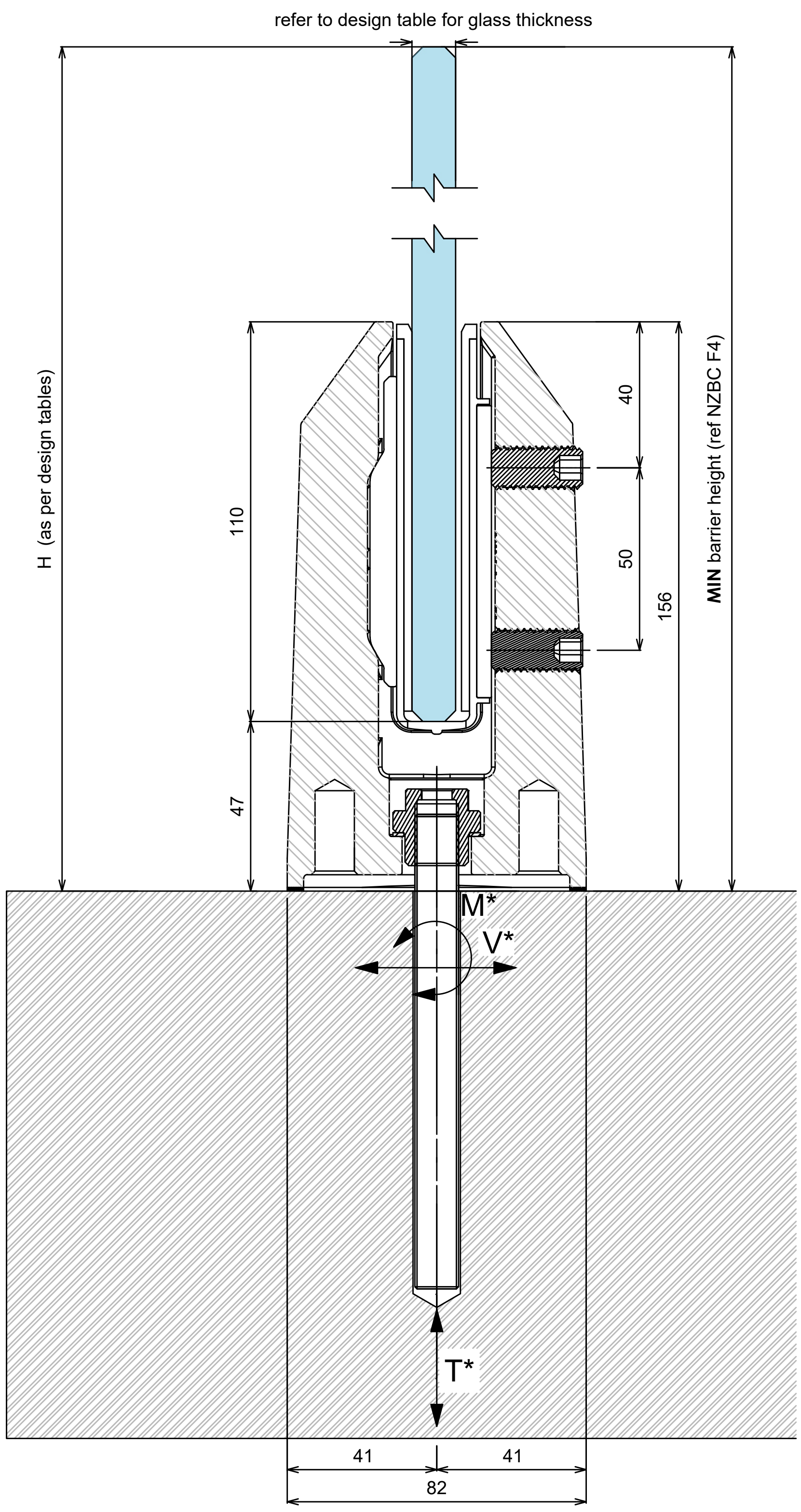
a. 12mm Toughened Safety Glass with interlinking tail
b. 13.52mm Toughened Laminated SentryGlas
c. 15mm Toughened Safety Glass with interlinking tail
d. 17.52mm Toughened Laminated SentryGlas

NOTE: All 12mm & 15mm glass balustrade to be provided with an interlinking rail as per NZBC NZS4223.3 and F4

DOUBLE FIX OPTION



SINGLE FIX OPTION



FOR RESIDENTIAL OCCUPANCY A, A OTHER & C3 OF TABLE 3.3 AS/NZS 1170.1

FOR PROTECTING A FALL OF MORE THAN 1M AND LESS THAN 5M FROM THE GROUND

Cantilever Glass Balustrade – Occupation A, C3,(Residential) to Table 3.3 of AS/NZS 1170.1:2002 and NZS4223.3:2016,

Occupancy	Glass type/ Thickness t (mm)	Maximum Design Height H above finish floor (mm)	Maximum design wind pressure		Maximum spigots internal spacing (mm)	Spigot edge/overhang distance (mm)	Substrate fixing		
			SLS Wind pressure (kPa)	ULS Wind Pressure (kPa)			Concrete/ Steel		Notes
							ULS M* kNm/spacing	ULS N* kN/spacing	
Residential & Domestic only Occupancy types A, and C3	12mm Toughened, 13.52 SentryGlas	1200	H /0.97	H/1.36	720	250	0.79	16	
		1100	VH /1.25	VH/1.76	720	250	0.87	15.5	
		1000	EH /1.51	EH/2.13	720	250	0.90	15	
	15mm Toughened, 17.52mm SentryGlas	1300	H /0.97	H/1.36	720	250	0.98	17.58	
		1250	VH /1.25	VH/1.76	720	250	0.97	17.32	
		1200	EH /1.51	EH/2.13	720	250	0.93	16.5	

TABLE 02 Residential top Fix Balustrade using SP14 Spigots – Protection from Falling of more than 1m and less than 5m (Timber substrate using THRU BOLTS DOUBLE FIXING ONLY)

Cantilever Glass Balustrade – Occupation A, C3,(Residential) to Table 3.3 of AS/NZS 1170.1:2002 and NZS4223.3:2016,

Occupancy	Glass type/ Thickness t (mm)	Maximum Design Height H above finish floor (mm)	Maximum design wind pressure		Maximum spigots internal spacing (mm)	Spigot edge/overhang distance (mm)	Substrate fixing		
			SLS Wind pressure (kPa)	ULS Wind Pressure (kPa)			Concrete/ Steel		Notes
							ULS M* kNm/spacing	ULS N* kN/spacing	
Residential & Domestic only Occupancy types A, and C3	12mm Toughened, 13.52 SentryGlas	1100	H /0.97	H/1.36	600	250	0.79	16.88	
		1100	VH /1.25	VH/1.76	500	250	0.87	15.5	

TABLE 01 Residential Pool Top Fix Fence using SP14 Spigots – NOT Protecting from Falling of more than 1m (Minimum fence heights =1200mm) Concrete and steel substrates Single/double fixing

Glass type/ Thickness t (mm)	Maximum design wind pressure		Maximum Design Height H above finish floor (mm)	Maximum spigots internal spacing (mm)	Spigot edge distance (mm)	Substrate fixing		
						Concrete/Steel		
	SLS Wind pressure (kPa)	ULS Wind Pressure (kPa)				ULS M* kNm/spacing	ULS N* kN/spacing	Notes
12mm T, 13.52mm SG	M/0.68	M/0.96	1300	1000	500	0.68	12.15	
	H/0.97	H/1.36	1200	1000	500	0.75	13.8	
	VH/1.25	VH/1.76	1200	750	375	0.83	14.84	
15mm T, 17.52mm SG	H/0.97	H/1.36	1300	1000	500	1.12	20	
	VH/1.25	VH/1.76	1300	750	375	1.13	20	
	EH/1.51	EH/2.13	1300	600	300	1.17	20	

TABLE 02 Residential Pool Top Fix Fence using SP14 Spigots – NOT Protecting from Falling of more than 1m (Minimum fence heights =1200mm) timber substrates using SS Coach screws/SS threaded rod with double fixing:

Glass type/ Thickness t (mm)	Maximum design wind pressure		Maximum Design Height H above finish floor (mm)	Maximum spigots internal spacing (mm)	Spigot edge distance (mm)	Substrate fixing		
						Timber		
	SLS Wind pressure (kPa)	ULS Wind Pressure (kPa)				ULS M* kNm/spacing	ULS N* kN/spacing	Notes
12mm T, 13.52mm SG	M/0.68	M/0.96	1200	1000	500	0.6	10.8	
	H/0.97	H/1.36	1200	750	375	0.75	13.38	Can be fixed only to hySPAN joists with 160mm minimum penetration into timber
	VH/1.25	VH/1.76	1200	600	300	0.83	14.84	Can be fixed only to hySPAN joists with 160mm minimum penetration into timber

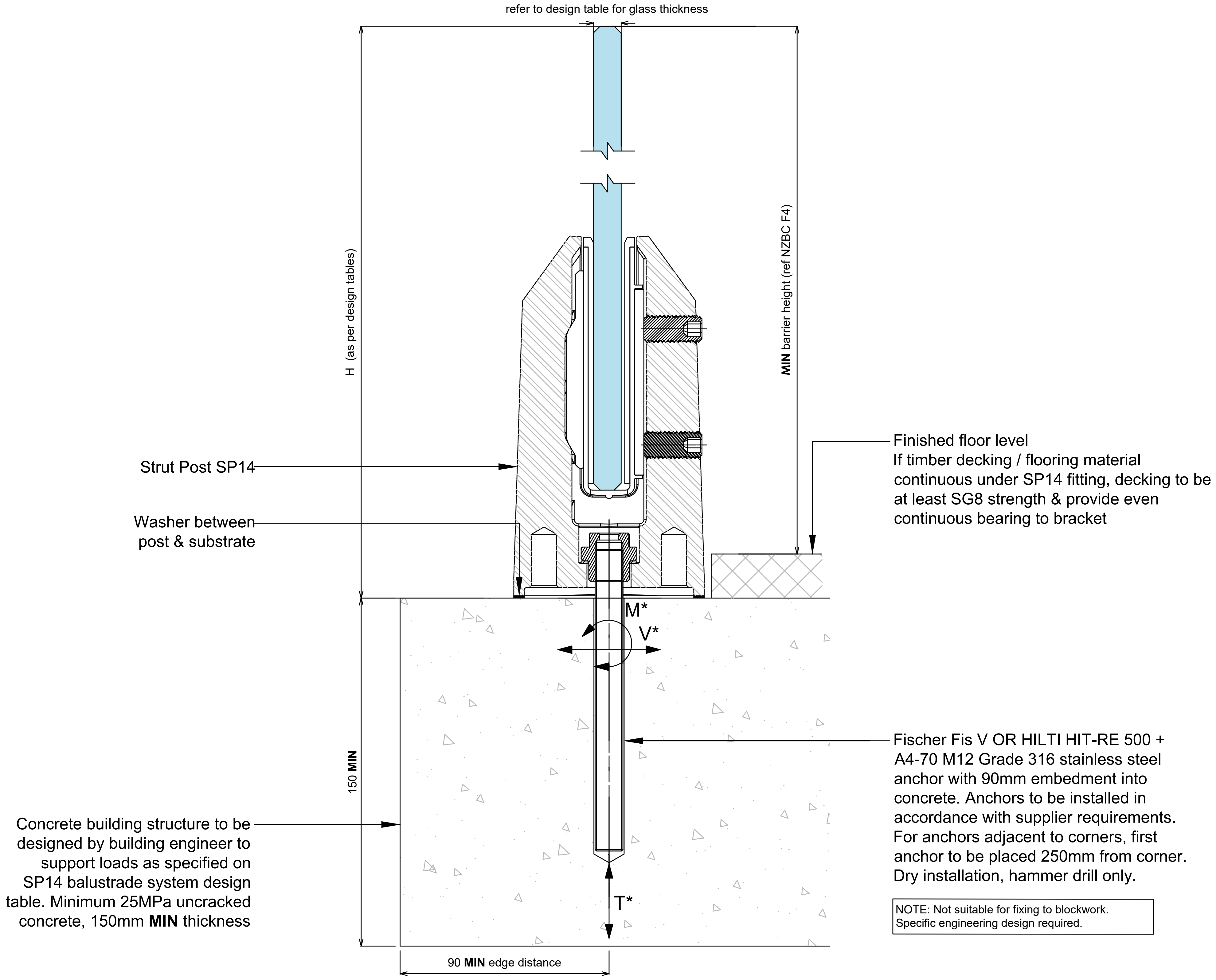
1) Refer to elevation drawings for Height 'H'.
2) The specifier must ensure the balustrade height above floor level requirements as per the NZ Building Code are complied with.
3) Design loads are in accordance with AS/NZS 1170.1:2002 table 3.3 as modified by NZBC B1/VM1 and DBH Guidance on Barrier Design (March 2012).
4) Capacity of all substructure is to be verified by building engineer or checked for accordance with NZS3604 (where applicable) prior to fixing.
5) Refer to the relevant fixing details on drawings
6) For designs outside the scope of these tables and ULS wind pressures exceeding those shown, specific design is required.
7) Maximum 10mm tolerance allowed to H heights noted in table.
8) SP14 STRUT POST cover does not posess toe hold. Only certified glass with stamp to be used with both balustrade and pool fence system

FOR RESIDENTIAL OCCUPANCY A, A OTHER & C3 OF TABLE 3.3 AS/NZS 1170.1

FOR PROTECTING A FALL OF MORE THAN 1M AND LESS THAN 5M FROM THE GROUND

Refer to SP14 balustrade system design table for required glass thickness, fixing spacings and fixing loads according to AS/NZS 1170.1:2002 for the occupancies listed above.

NOTES: Refer to design tables and elevations for post failure requirements. Interlinking rail / clips not shown for clarity. 'H' refers to top of barrier.



Notes:

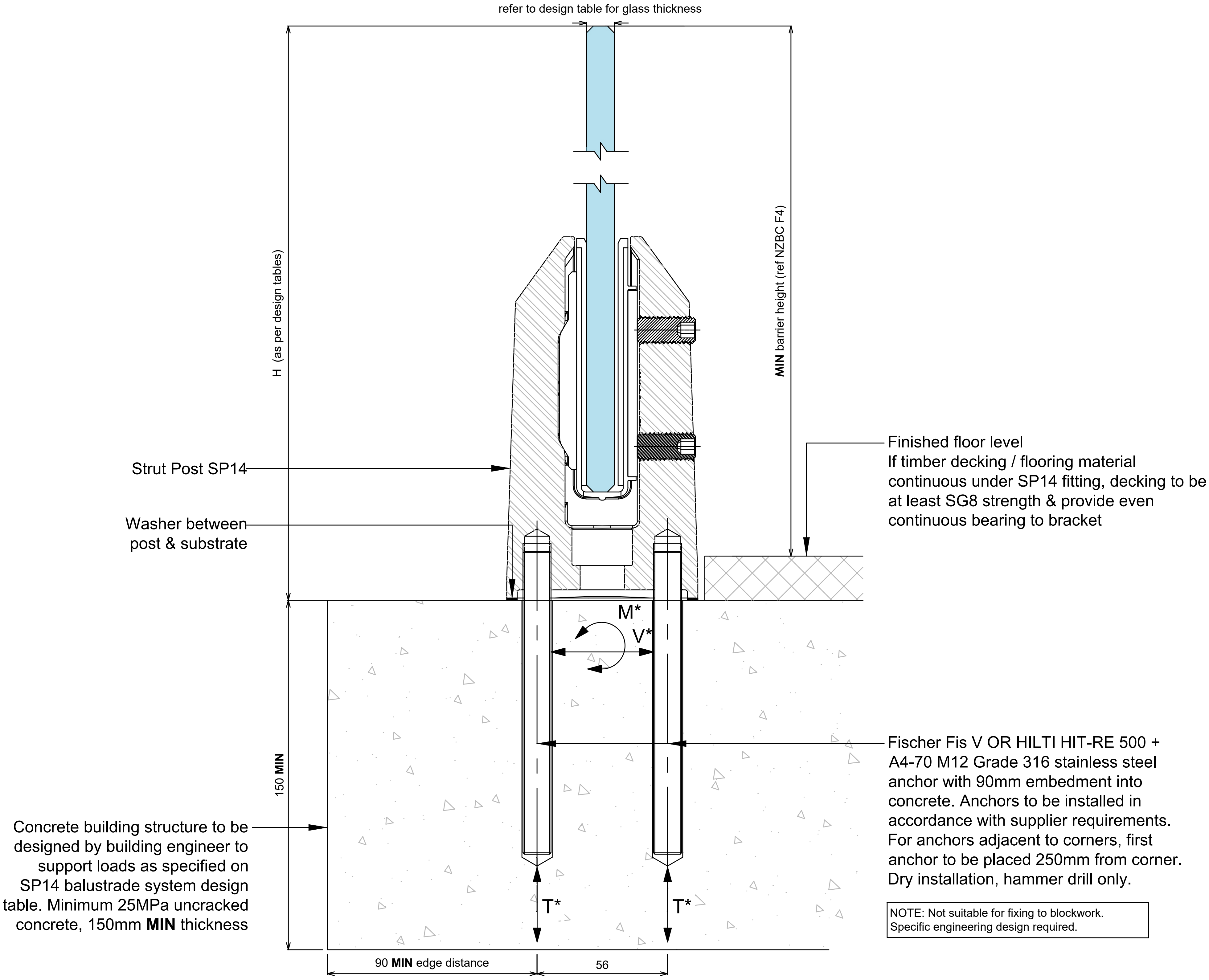
1) Capacity of structure is to be of sufficient strength to support loads M*, V* and T* specified on SP14 Strut Post balustrade system design table. Structure capacity to be verified by building engineer prior to fixing balustrade.
2) Max loading to comply with AS/NZS 1170.1:2002 Minimum Imposed Actions for Barriers Occupancy, shown at top of drawing, for design in accordance with SP14 Strut Post balustrade system design table.
3) Penetration through a membrane requires specific engineers design
4) No substitution allowed - any variation from the details above and design tables will require specific design.
5) Substructure designer to ensure that there is no water runoff from dissimilar metals or treated timber onto barrier components.

FOR RESIDENTIAL OCCUPANCY A, A OTHER & C3 OF TABLE 3.3 AS/NZS 1170.1

FOR PROTECTING A FALL OF MORE THAN 1M AND LESS THAN 5M FROM THE GROUND

Refer to SP14 balustrade system design table for required glass thickness, fixing spacings and fixing loads according to AS/NZS 1170.1:2002 for the occupancies listed above.

NOTES: Refer to design tables and elevations for post failure requirements. Interlinking rail / clips not shown for clarity. 'H' refers to top of barrier.



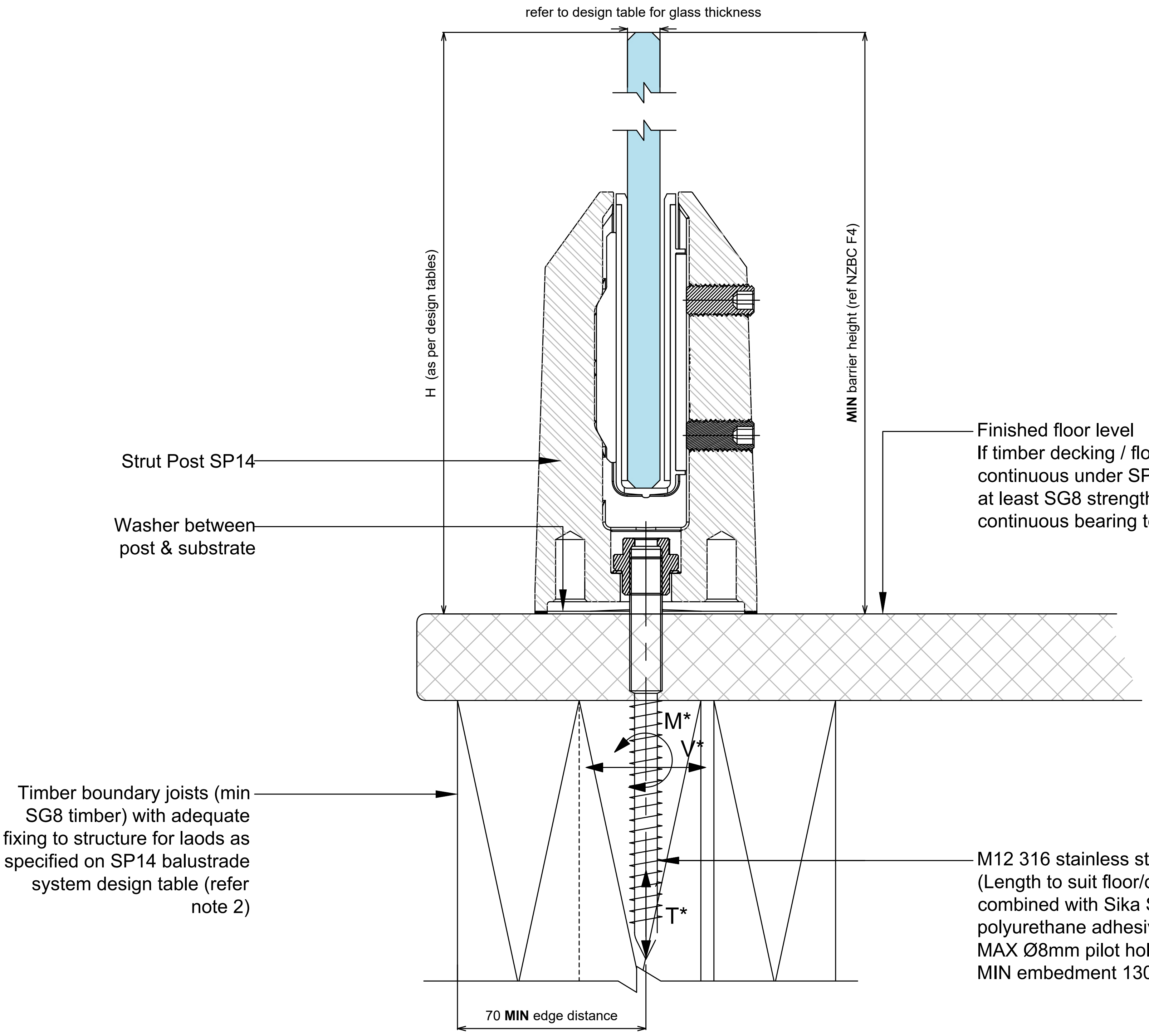
Notes:

1) Capacity of structure is to be of sufficient strength to support loads M*, V* and T* specified on SP14 Strut Post balustrade system design table. Structure capacity to be verified by building engineer prior to fixing balustrade.
2) Max loading to comply with AS/NZS 1170.1:2002 Minimum Imposed Actions for Barriers Occupancy, shown at top of drawing, for design in accordance with SP14 Strut Post balustrade system design table.
3) Penetration through a membrane requires specific engineers design
4) No substitution allowed - any variation from the details above and design tables will require specific design.
5) Substructure designer to ensure that there is no water runoff from dissimilar metals or treated timber onto barrier components.

FOR POOL FENCES ONLY

Refer to SP14 balustrade system design table for required glass thickness, fixing spacings and fixing loads according to AS/NZS 1170.1:2002 for the occupancies listed above.

NOTES: Refer to design tables and elevations for post ~~fai~~lure requirements. Interlinking rail / clips not shown for clarity. 'H' refers to top of barrier.



NOTES:

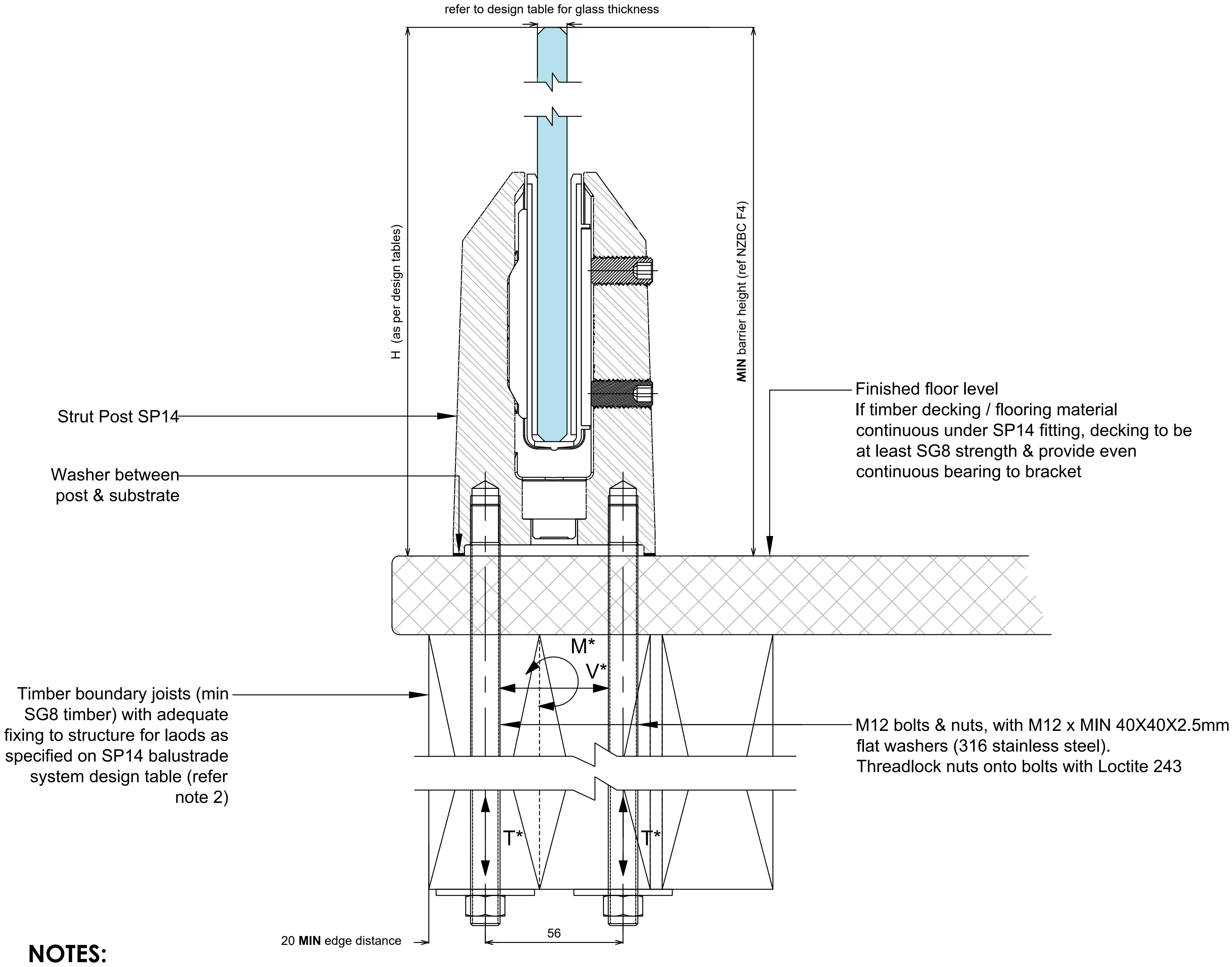
1) Capacity of structure is to be of sufficient strength to support loads M*, V*, T* and torsion of joints specified on SP14 Strut Post balustrade system design table. Structure capacity to be verified by building engineer, or where applicable checked to NZS3604 requirements prior to fixing balustrade.
2) Timber decks designed to NZS 3604:2011 guidelines will meet loading requirement. Additional blocking required for inner fixing.
3) Max loading to comply with AS/NZS 1170.1:2002 Minimum Imposed Actions for Barriers Occupancy, shown at top of drawing, for design in accordance with SP14 Strut Post balustrade system design table.
4) Penetration through a membrane requires specific engineering design
5) For fixing to timber substrates, the installer shall ensure that the bolt / coach screw is sufficiently tightened to reduce movement of the bolt head and washer. Care should be taken not to over tighten the fixings that would cause crushing of the timber or compromise the thread leading to anchor pull-out.
6) No substitution allowed - any variation from the details above and design tables will require specific design.
7) Fixings to timber must be re-tightened 2 months after installation and periodically thereafter to allow for timber shrinkage.
8) Substructure designer to ensure that there is no water runoff from dissimilar metals or treated timber onto barrier components.

FOR RESIDENTIAL OCCUPANCY A, A OTHER & C3 OF TABLE 3.3 AS/NZS 1170.1

FOR PROTECTING A FALL OF MORE THAN 1M AND LESS THAN 5M FROM THE GROUND

Refer to SP14 balustrade system design table for required glass thickness, fixing spacings and fixing loads according to AS/NZS 1170.1:2002 for the occupancies listed above.

NOTES: Refer to design tables and elevations for post failure requirements. Interlinking rail / clips not shown for clarity. 'H' refers to top of barrier.



NOTES:

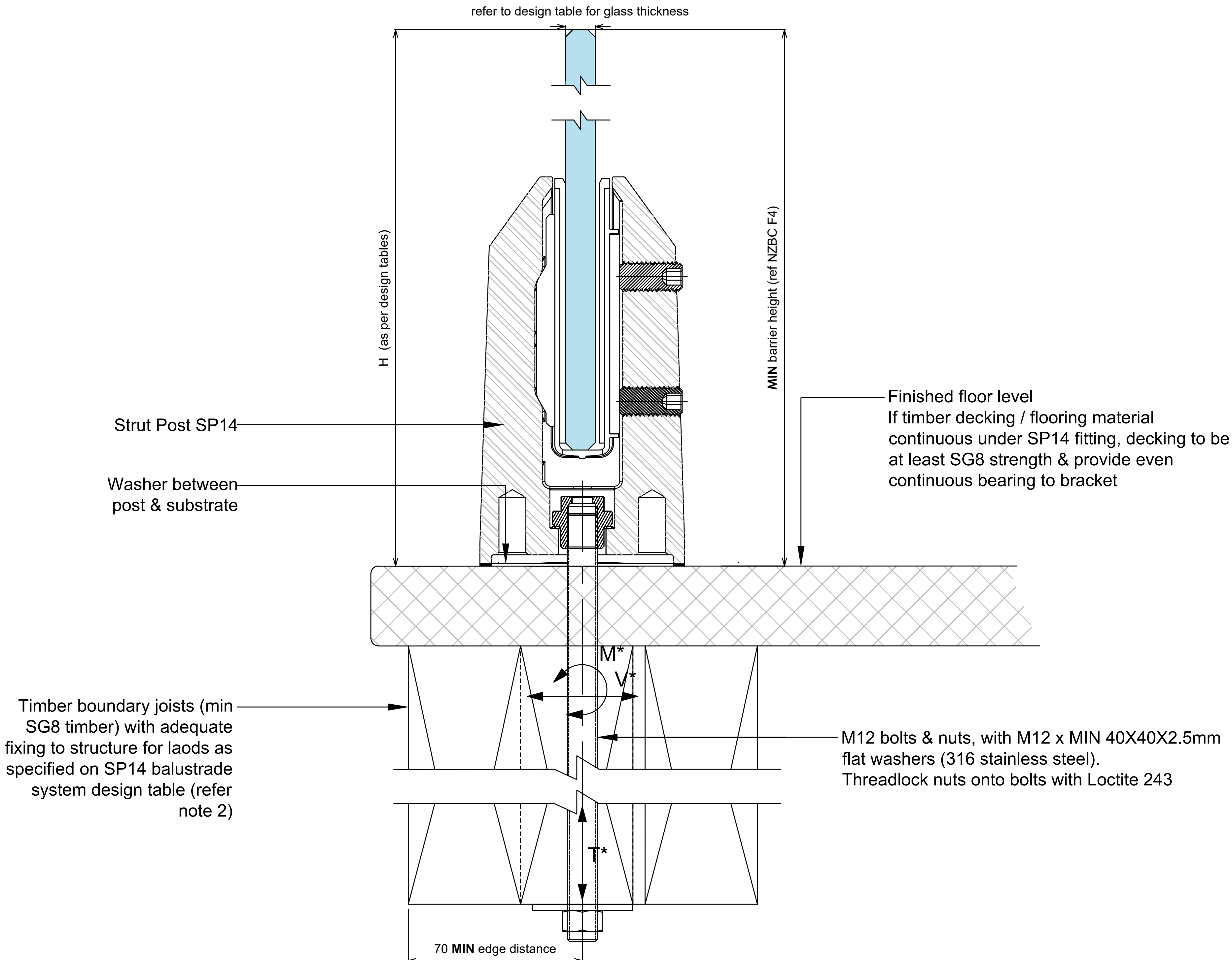
1) Capacity of structure is to be of sufficient strength to support loads M*, V*, T* and torsion of joints specified on SP14 Strut Post balustrade system design table. Structure capacity to be verified by building engineer, or where applicable checked to NZS3604 requirements prior to fixing balustrade.
2) Timber decks designed to NZS 3604:2011 guidelines will meet loading requirement. Additional blocking required for inner fixing.
3) Max loading to comply with AS/NZS 1170.1:2002 Minimum Imposed Actions for Barriers Occupancy, shown at top of drawing, for design in accordance with SP14 Strut Post balustrade system design table.
4) Penetration through a membrane requires specific engineering design
5) For fixing to timber substrates, the installer shall ensure that the bolt / coach screw is sufficiently tightened to reduce movement of the bolt head and washer. Care should be taken not to over tighten the fixings that would cause crushing of the timber or compromise the thread leading to anchor pull-out.
6) No substitution allowed - any variation from the details above and design tables will require specific design.
7) Fixings to timber must be re-tightened 2 months after installation and periodically thereafter to allow for timber shrinkage.
8) Substructure designer to ensure that there is no water runoff from dissimilar metals or treated timber onto barrier components.

FOR RESIDENTIAL OCCUPANCY A, A OTHER & C3 OF TABLE 3.3 AS/NZS 1170.1

FOR PROTECTING A FALL OF MORE THAN 1M AND LESS THAN 5M FROM THE GROUND

Refer to SP14 balustrade system design table for required glass thickness, fixing spacings and fixing loads according to AS/NZS 1170.1:2002 for the occupancies listed above.

NOTES: Refer to design tables and elevations for post failure requirements. Interlinking rail / clips not shown for clarity. 'H' refers to top of barrier.



NOTES:

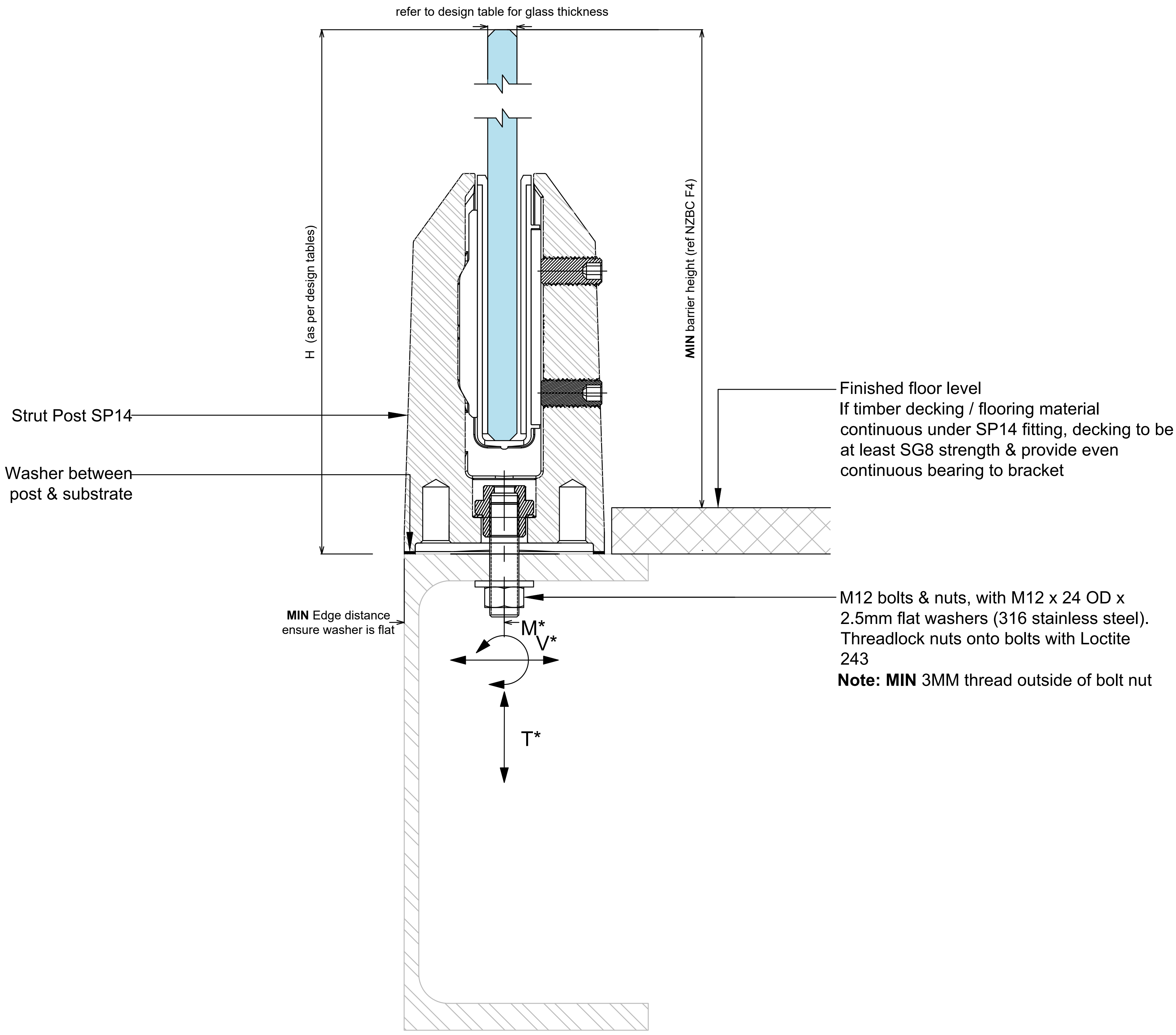
1) Capacity of structure is to be of sufficient strength to support loads M*, V*, T* and torsion of joints specified on SP14 Strut Post balustrade system design table. Structure capacity to be verified by building engineer, or where applicable checked to NZS3604 requirements prior to fixing balustrade.
2) Timber decks designed to NZS 3604:2011 guidelines will meet loading requirement. Additional blocking required for inner fixing.
3) Max loading to comply with AS/NZS 1170.1:2002 Minimum Imposed Actions for Barriers Occupancy, shown at top of drawing, for design in accordance with SP14 Strut Post balustrade system design table.
4) Penetration through a membrane requires specific engineering design
5) For fixing to timber substrates, the installer shall ensure that the bolt / coach screw is sufficiently tightened to reduce movement of the bolt head and washer. Care should be taken not to over tighten the fixings that would cause crushing of the timber or compromise the thread leading to anchor pull-out.
6) No substitution allowed - any variation from the details above and design tables will require specific design.
7) Fixings to timber must be re-tightened 2 months after installation and periodically thereafter to allow for timber shrinkage.
8) Substructure designer to ensure that there is no water runoff from dissimilar metals or treated timber onto barrier components.

FOR RESIDENTIAL OCCUPANCY A, A OTHER & C3 OF TABLE 3.3 AS/NZS 1170.1

FOR PROTECTING A FALL OF MORE THAN 1M AND LESS THAN 5M FROM THE GROUND

Refer to SP14 balustrade system design table for required glass thickness, fixing spacings and fixing loads according to AS/NZS 1170.1:2002 for the occupancies listed above.

NOTES: Refer to design tables and elevations for post failure requirements. Interlinking rail / clips not shown for clarity. 'H' refers to top of barrier.

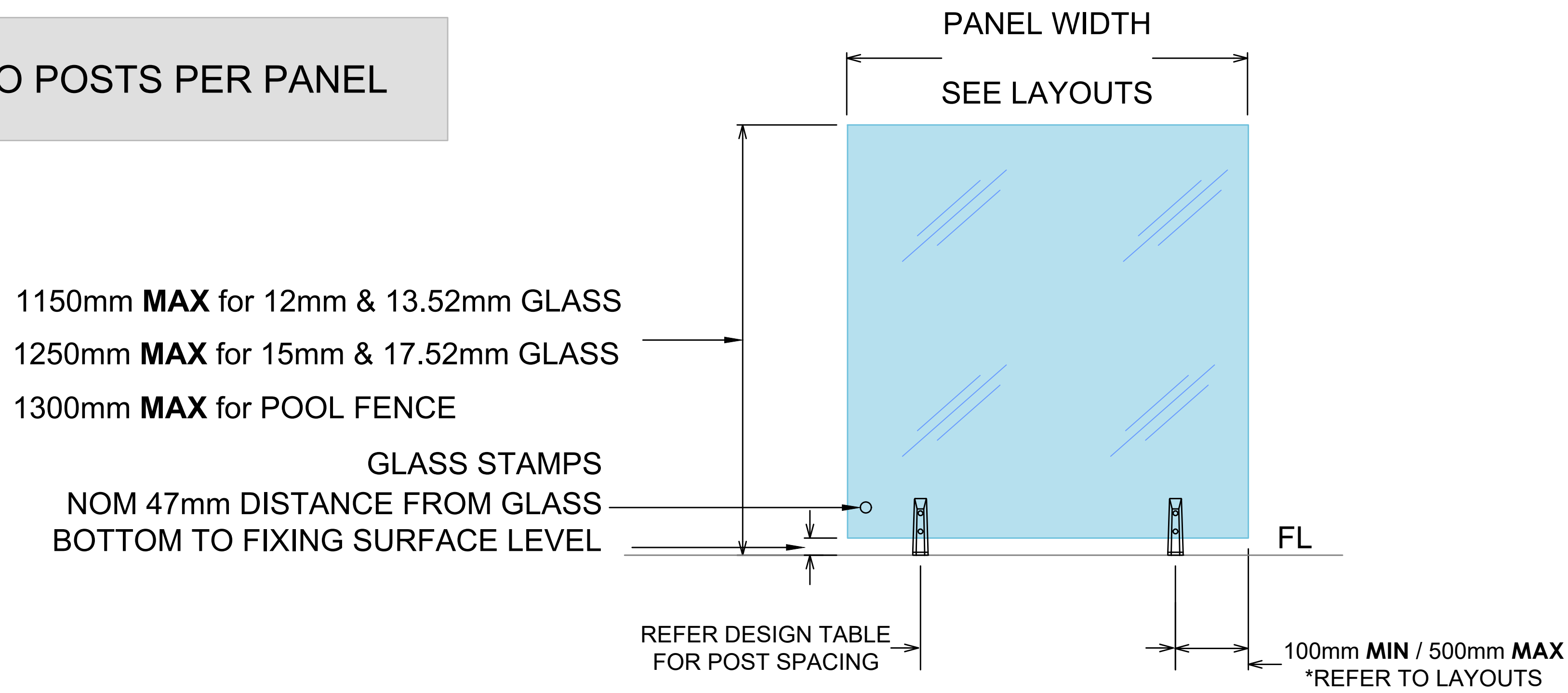


Notes:

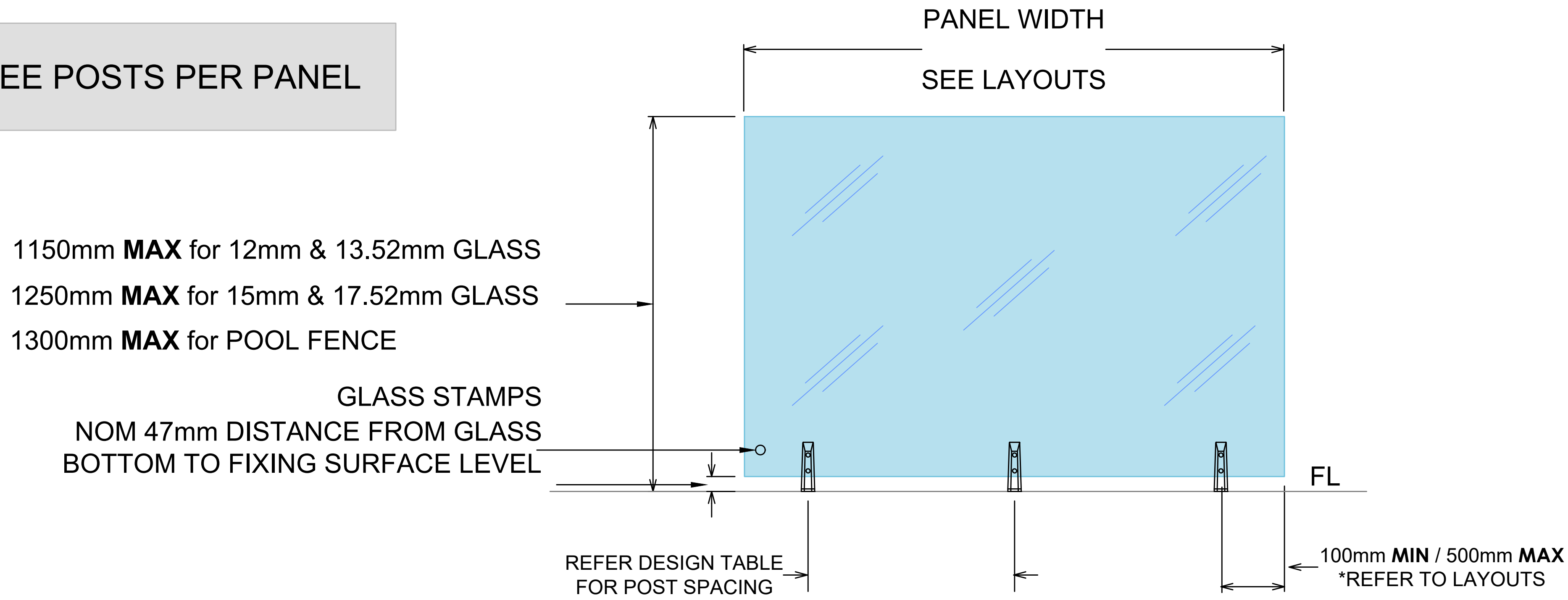
1) Capacity of structure is to be of sufficient strength to support loads M*, V* and T* specified on SP14 Strut Post balustrade system design table. Structure capacity to be verified by building engineer prior to fixing balustrade.
2) Max loading to comply with AS/NZS 1170.1:2002 Minimum Imposed Actions for Barriers Occupancy, shown at top of drawing, for design in accordance with SP14 Strut Post balustrade system design table.
3) Penetration through a membrane requires specific engineering design
4) For fixing to steel substrates, the installer shall ensure the bolts are tightened to a "snug-tight" level as defined in NZS3404.
5) No substitution allowed - any variation from the details above and design tables will require specific design.
6) Substructure designer to ensure that there is no water runoff from dissimilar metals or treated timber onto barrier components.

SP14 STRUT POST ELEVATION
TYPICAL POST SPACING LIMITS

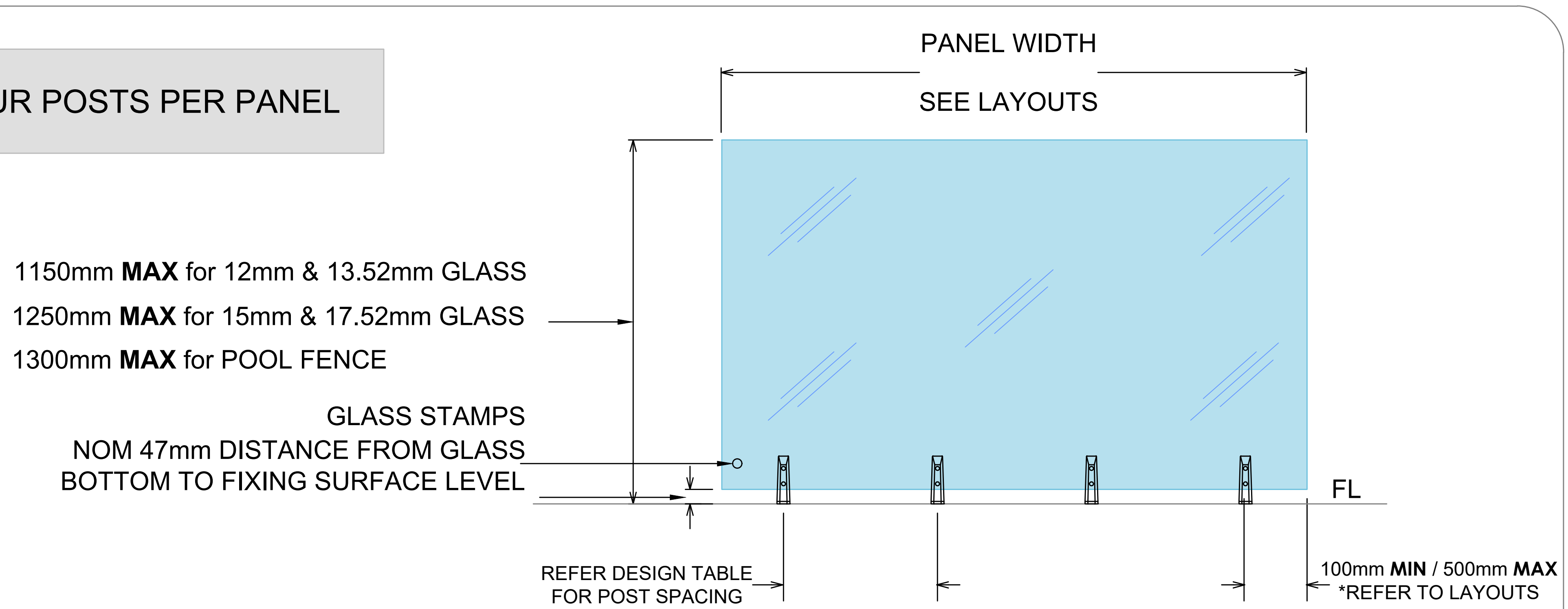
TWO POSTS PER PANEL



THREE POSTS PER PANEL

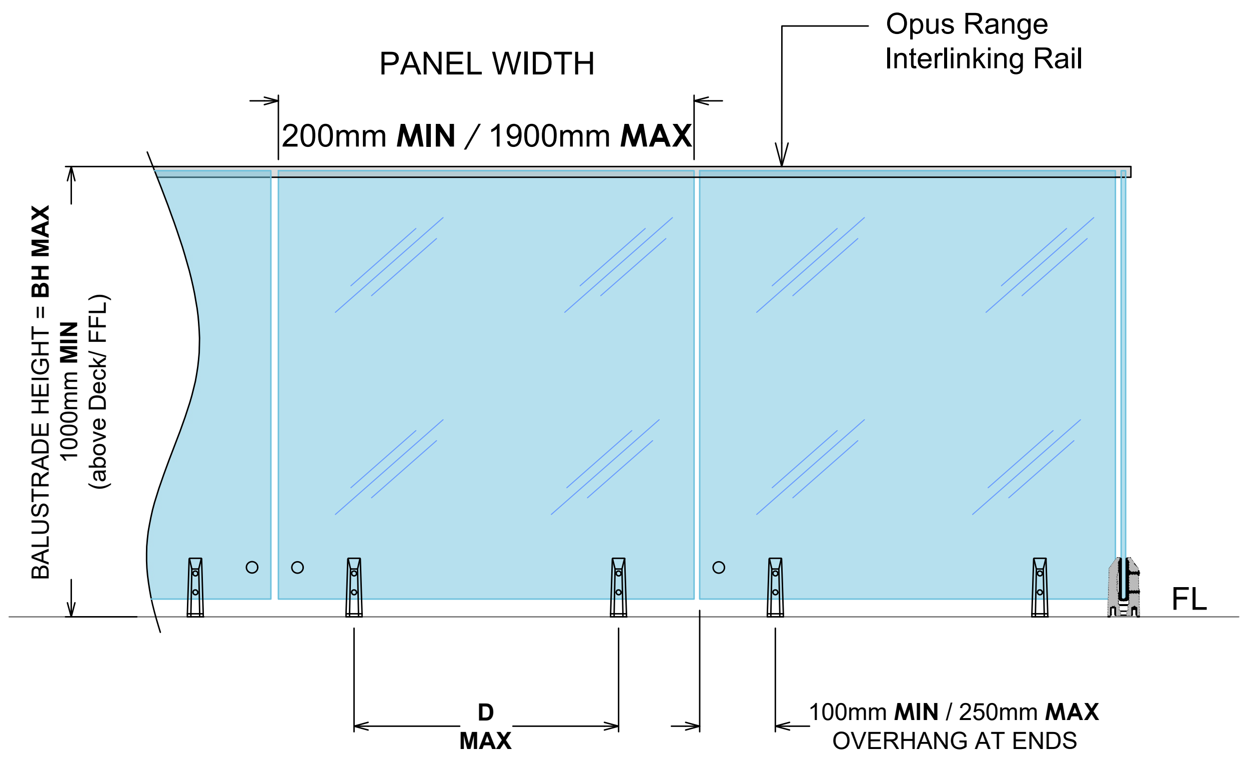


FOUR POSTS PER PANEL



STRUT POST SYSTEM 12mm TOUGHENED TYPICAL LAYOUT WITH INTERLINKING RAIL

Residential & domestic only Occupancy types A, A Other & C3	
12mm Toughened Glass	
HWZ	-BH MAX 1200mm -D MAX 720mm
VHWZ	-BH MAX 1100mm -D MAX 720mm
EHWZ	-BH MAX 1000mm -D MAX 720mm
MAX Tension Load per face fixing = 20kN	

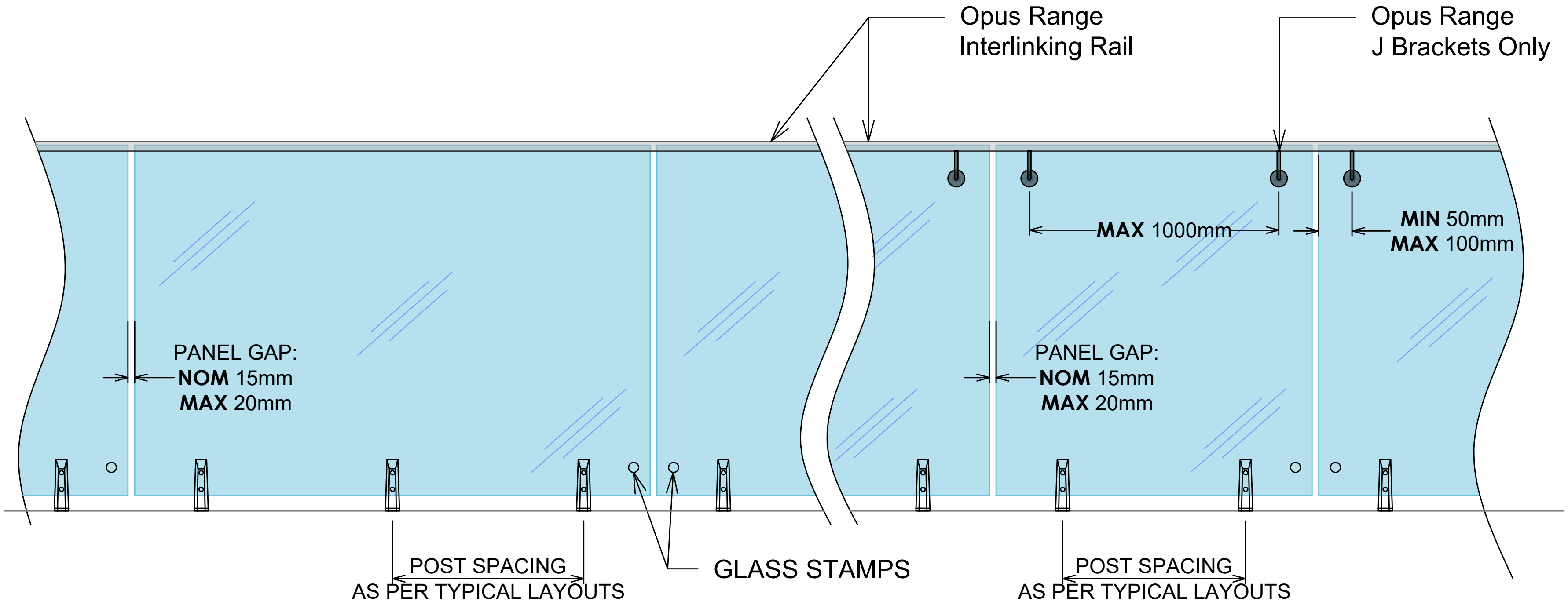


IMPORTANT NOTE: Exceeds the wind loadings for all Wind Zones up to and including **EXTRA HIGH WIND ZONE** as set out in NZS 3604:2011. Refer to the interlinking rail pages for conformance to NZS 4223.3.2016

STRUT POST SYSTEM 12mm TOUGHENEDTYPICAL MOUNTING OPTIONS

GLASS & FIXING SPECIFICATIONS:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

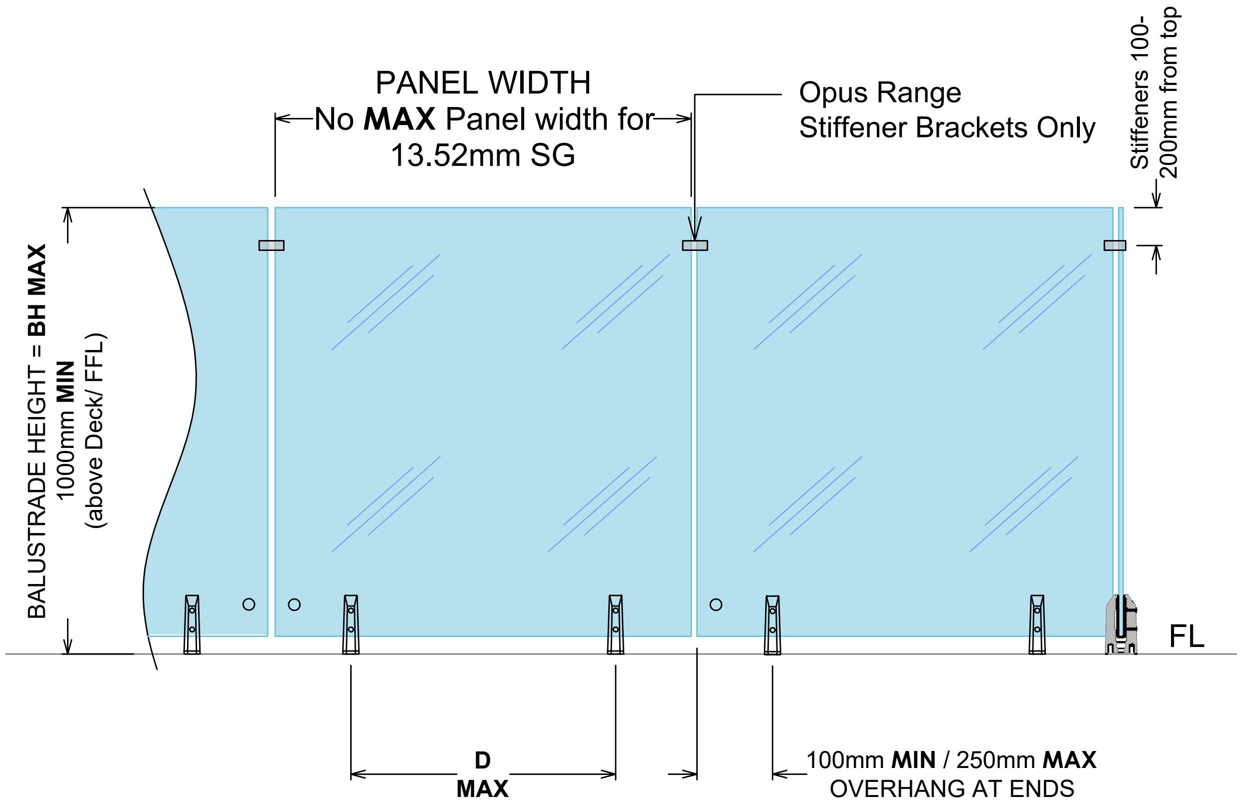
NOTE: Glass must have a **MIN** strength of 100MPa. All edges polished



IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.

STRUT POST SYSTEM 13.52mm SentryGlas TOUGHENED LAMINATE TYPICAL LAYOUT WITH STIFFENER BRACKETS

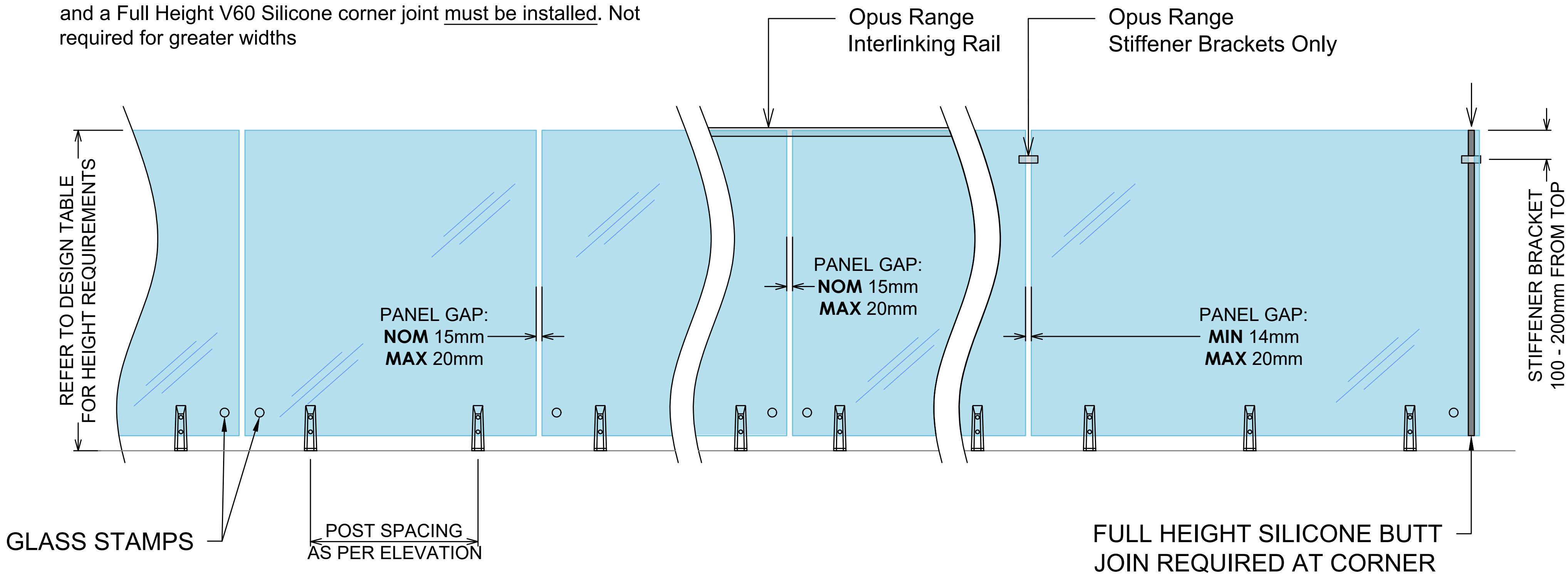
Residential & domestic only Occupancy types A, A Other & C3	
13.52mm SentryGlas	
HWZ	-BH MAX 1200mm -D MAX 720mm
VHWZ	-BH MAX 1100mm -D MAX 720mm
EHWZ	-BH MAX 1000mm -D MAX 720mm
MAX Tension Load per face fixing = 20kN	



IMPORTANT NOTE: Exceeds the wind loadings for all Wind Zones up to and including **EXTRA HIGH WIND ZONE** as set out in NZS 3604:2011. Refer to the stiffener bracket pages for conformance to NZS 4223.3.2016

STRUT POST SYSTEM 13.52mm SentryGlas TOUGHENED LAMINATE TYPICAL MOUNTING OPTIONS

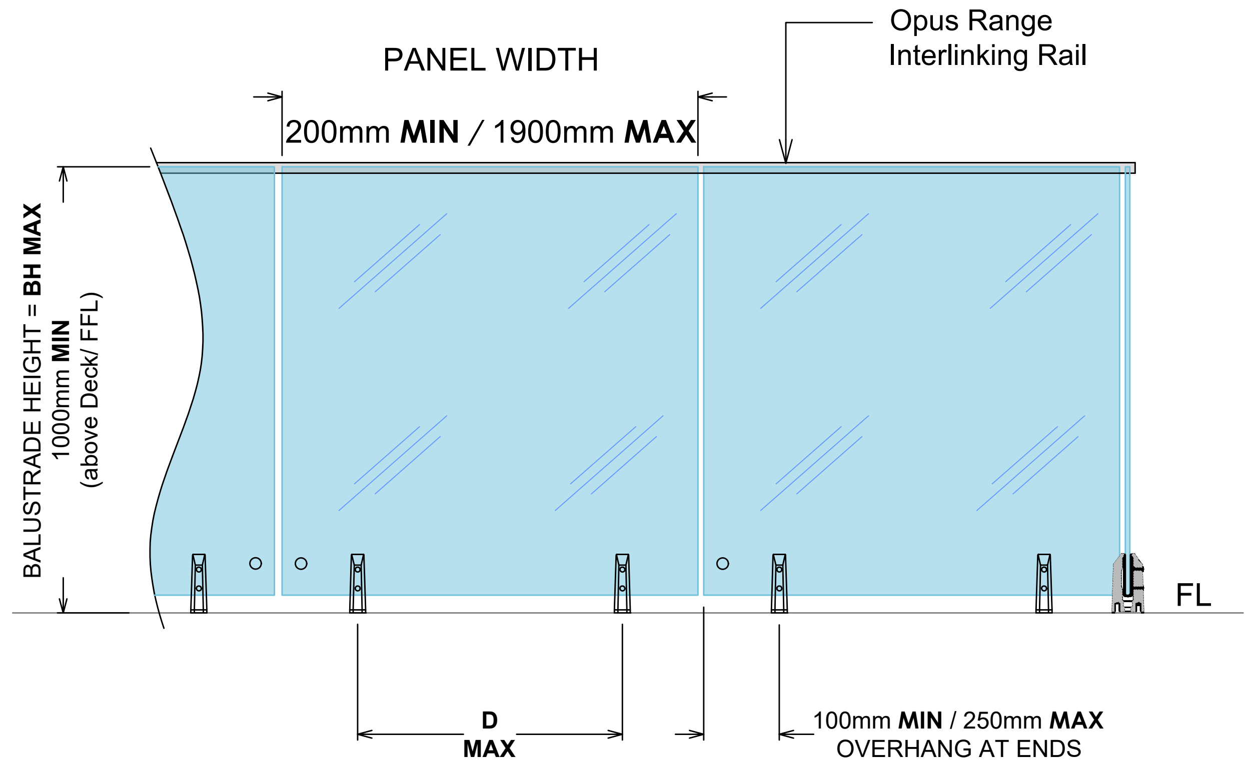
GLASS & FIXING SPECIFICATIONS:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.
NOTE: Glass must have a **MIN** strength of 100MPa. All edges polished
For 13.52 SentryGlas panel widths less than 1700mm Stiffener Brackets and a Full Height V60 Silicone corner joint must be installed. Not required for greater widths



IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.

STRUT POST SYSTEM 15mm TOUGHENED WITH INTERLINKING RAIL

Residential & domestic only Occupancy types A, A Other & C3	
15mm Toughened Glass	
HWZ	-BH MAX 1300mm -D MAX 720mm
VHWZ	-BH MAX 1250mm -D MAX 720mm
EHWZ	-BH MAX 1200mm -D MAX 720mm
MAX Tension Load per face fixing = 20kN	

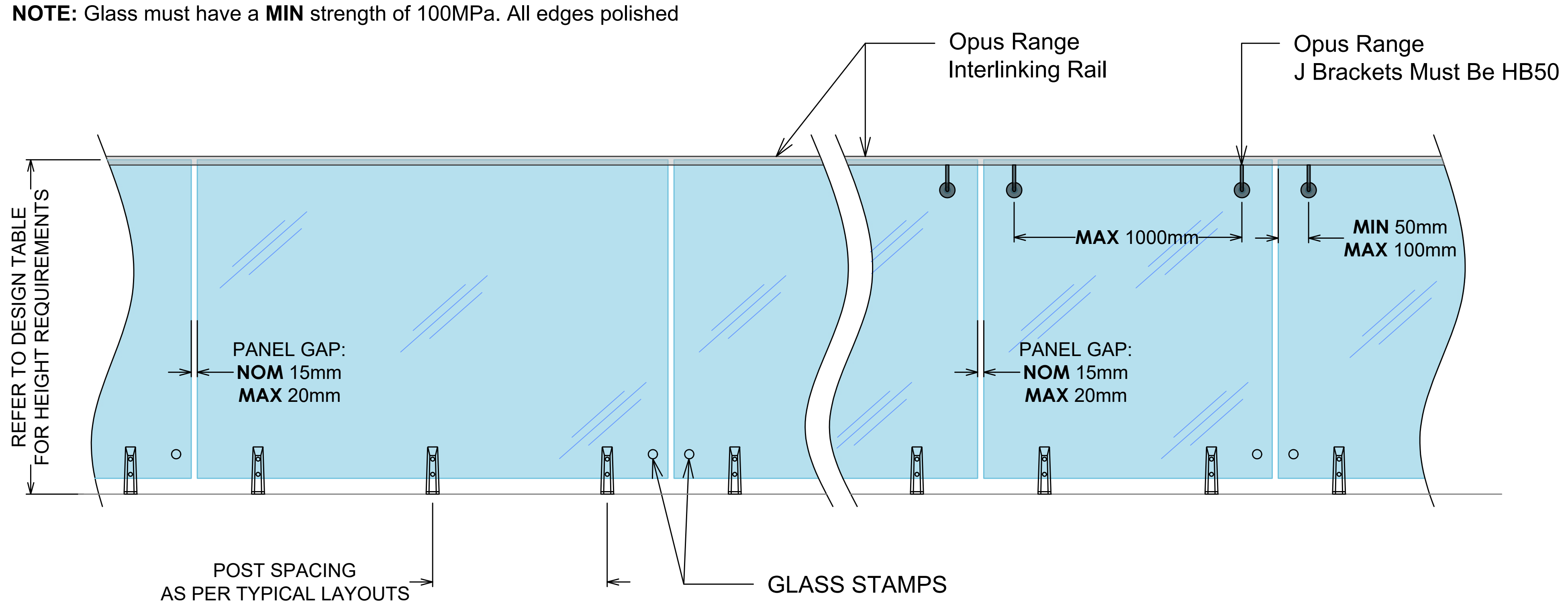


IMPORTANT NOTE: Exceeds the wind loadings for all Wind Zones up to and including **EXTRA HIGH WIND ZONE** as set out in NZS 3604:2011. Refer to the interlinking rail pages for conformance to NZS 4223.3.2016

STRUT POST SYSTEM 15mm TOUGHENED WITH INTERLINKING RAIL

GLASS & FIXING SPECIFICATIONS:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

NOTE: Glass must have a **MIN** strength of 100MPa. All edges polished



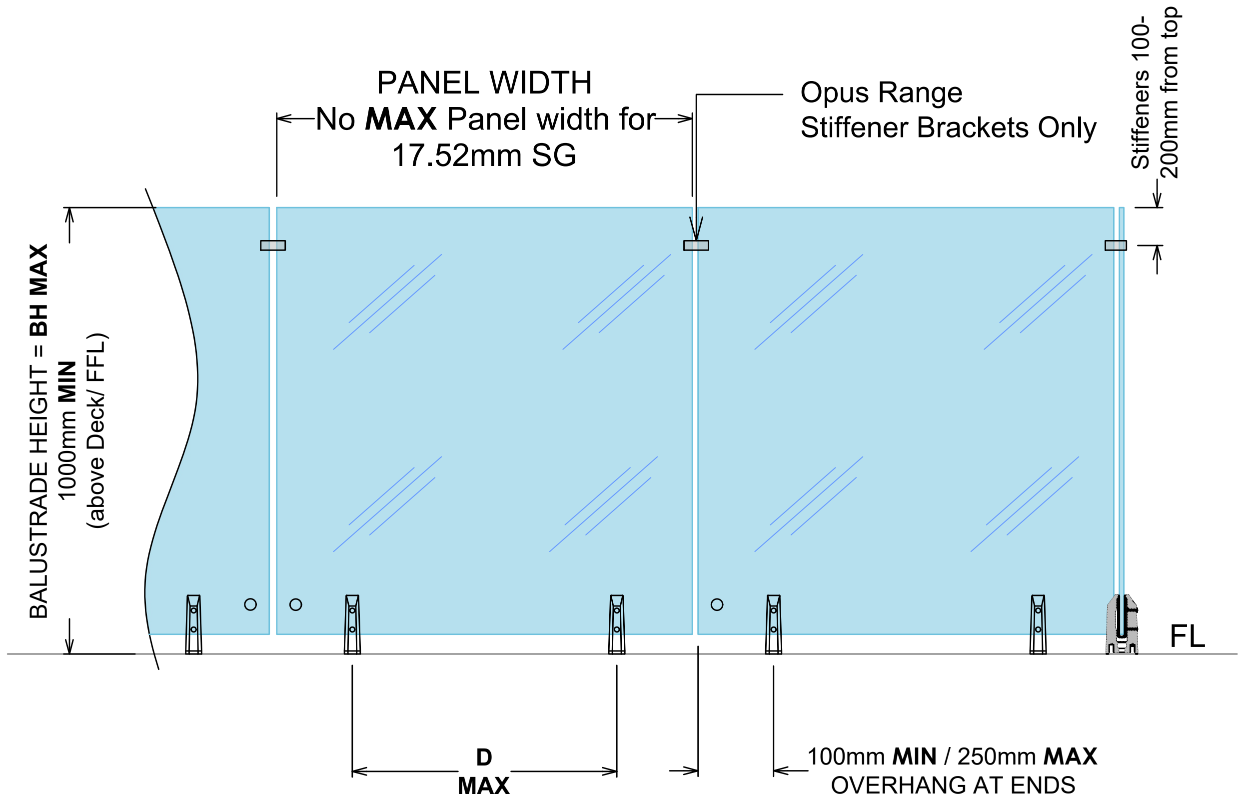
IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.

STRUT POST SYSTEM 17.52mm SentryGlas TOUGHENED LAMINATE TYPICAL LAYOUT WITH STIFFENER BRACKETS

Residential & domestic only
Occupancy types A, A Other & C3

17.52 SentryGlas	
HWZ	**-BH MAX 1300mm** **-D MAX 720mm**
VHWZ	**-BH MAX 1250mm** **-D MAX 720mm**
EHWZ	**-BH MAX 1200mm** **-D MAX 720mm**

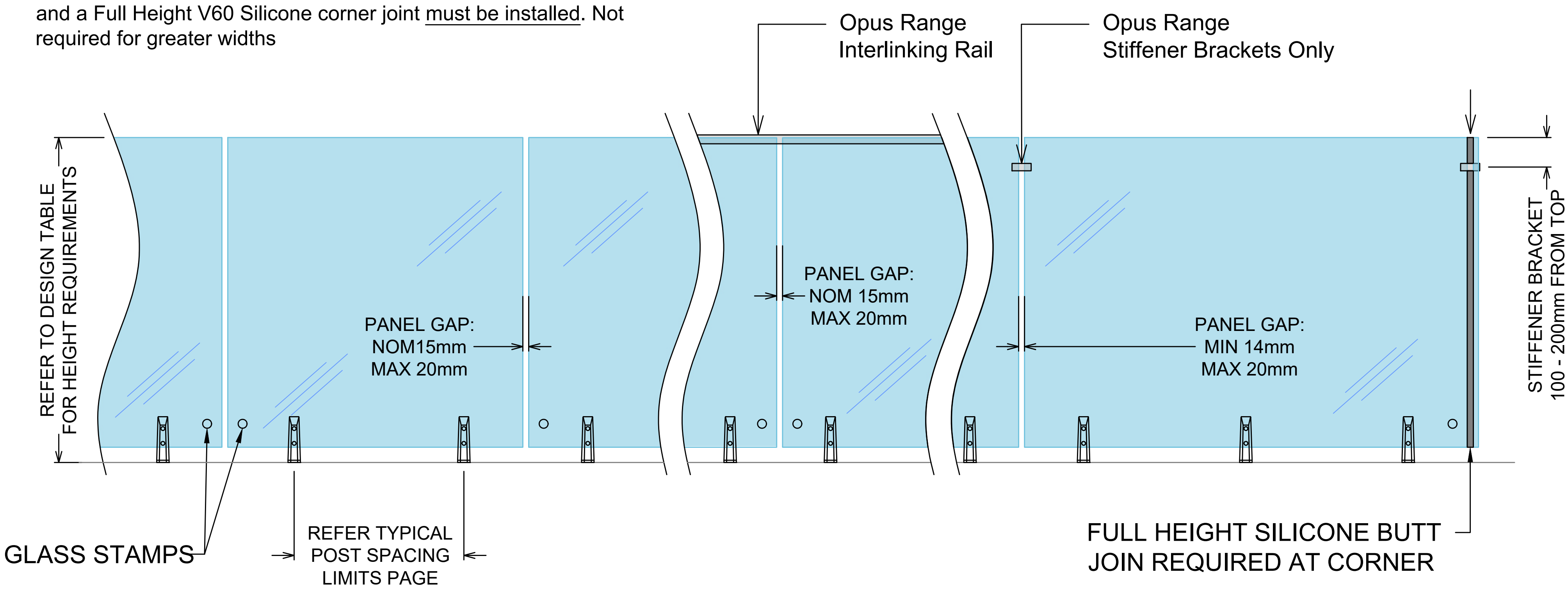
MAX Tension Load per face fixing = 23kN



IMPORTANT NOTE: Exceeds the wind loadings for all Wind Zones up to and including **EXTRA HIGH WIND ZONE** as set out in NZS 3604:2011. Refer to the stiffener bracket pages for conformance to NZS 4223.3.2016

STRUT POST SYSTEM 17.52mm SentryGlas TOUGHENED LAMINATE TYPICAL MOUNTING OPTIONS

GLASS & FIXING SPECIFICATIONS:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.
NOTE: Glass must have a **MIN** strength of 100MPa. All edges polished
For 17.52 SentryGlas panel widths less than 1300mm Stiffener Brackets and a Full Height V60 Silicone corner joint must be installed. Not required for greater widths



IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.

SP14 STRUT POST ELEVATION
TYPICAL LAYOUTS AND MOUNTING OPTIONS

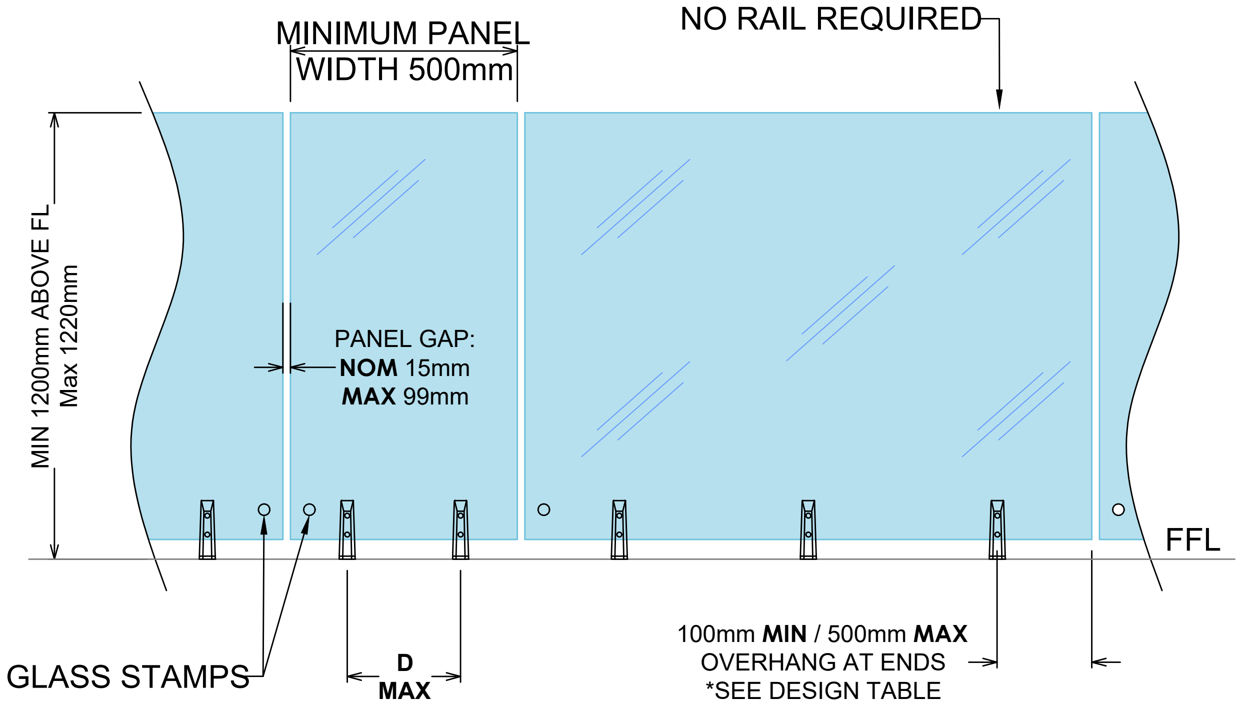
STRUT POST SYSTEM
POOL FENCE ONLY
12mm TOUGHENED

POOL FENCE	
12mm Toughened Glass	
MWZ	**-BH MAX 1200mm** **-D MAX 1200mm**
HWZ	**-BH MAX 1200mm** **-D MAX 1000mm**
VHWZ	**-BH MAX 1200mm** **-D MAX 750mm**
MAX Tension Load per face fixing = 23kN	

GLASS & FIXING SPECIFICATIONS:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

APPLIES TO FREE STANDING POOL FENCES NOT PROTECTING A FALL OF > 1000mm.

As of JAN 2017, complies with Building Code clause F9 & section 162C of the building act.



IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.

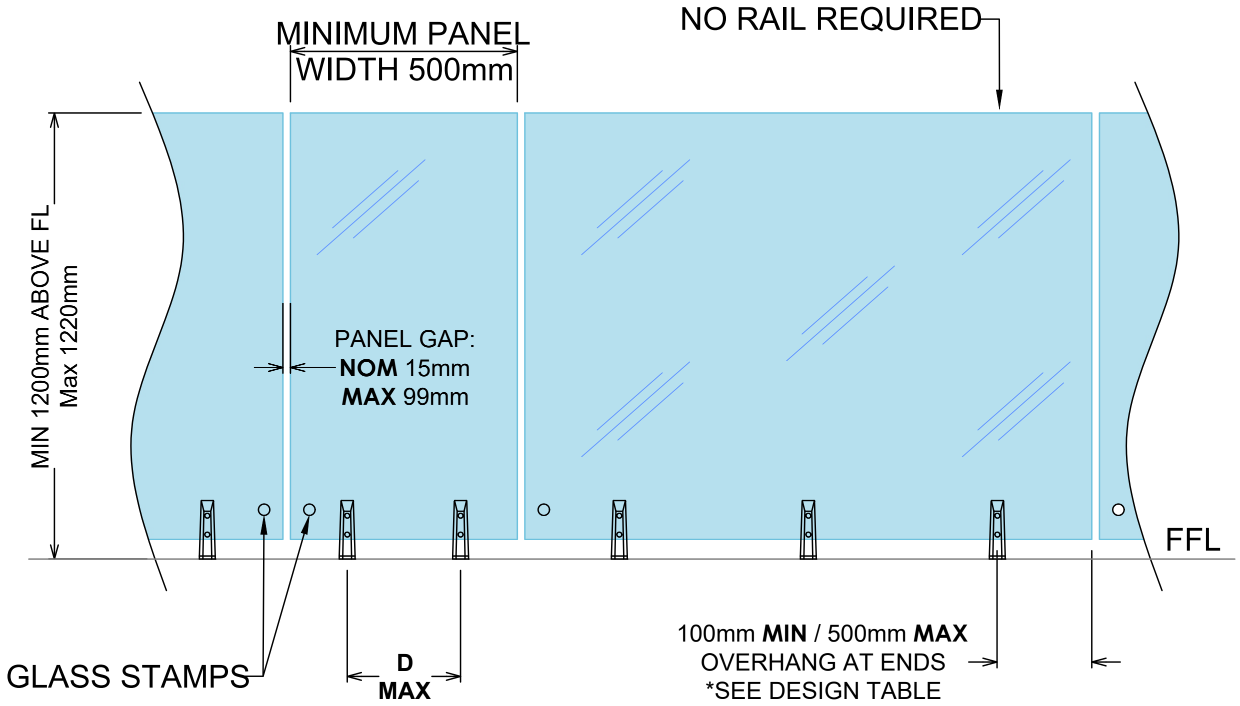
STRUT POST SYSTEM
POOL FENCE ONLY
15mm TOUGHENED

POOL FENCE	
15mm Toughened Glass	
MWZ	**-BH MAX 1300mm** **-D MAX 1000mm**
HWZ	**-BH MAX 1300mm** **-D MAX 750mm**
VHWZ	**-BH MAX 1300mm** **-D MAX 600mm**
MAX Tension Load per face fixing = 23kN	

GLASS & FIXING SPECIFICATIONS:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

APPLIES TO FREE STANDING POOL FENCES NOT PROTECTING A FALL OF > 1000mm.

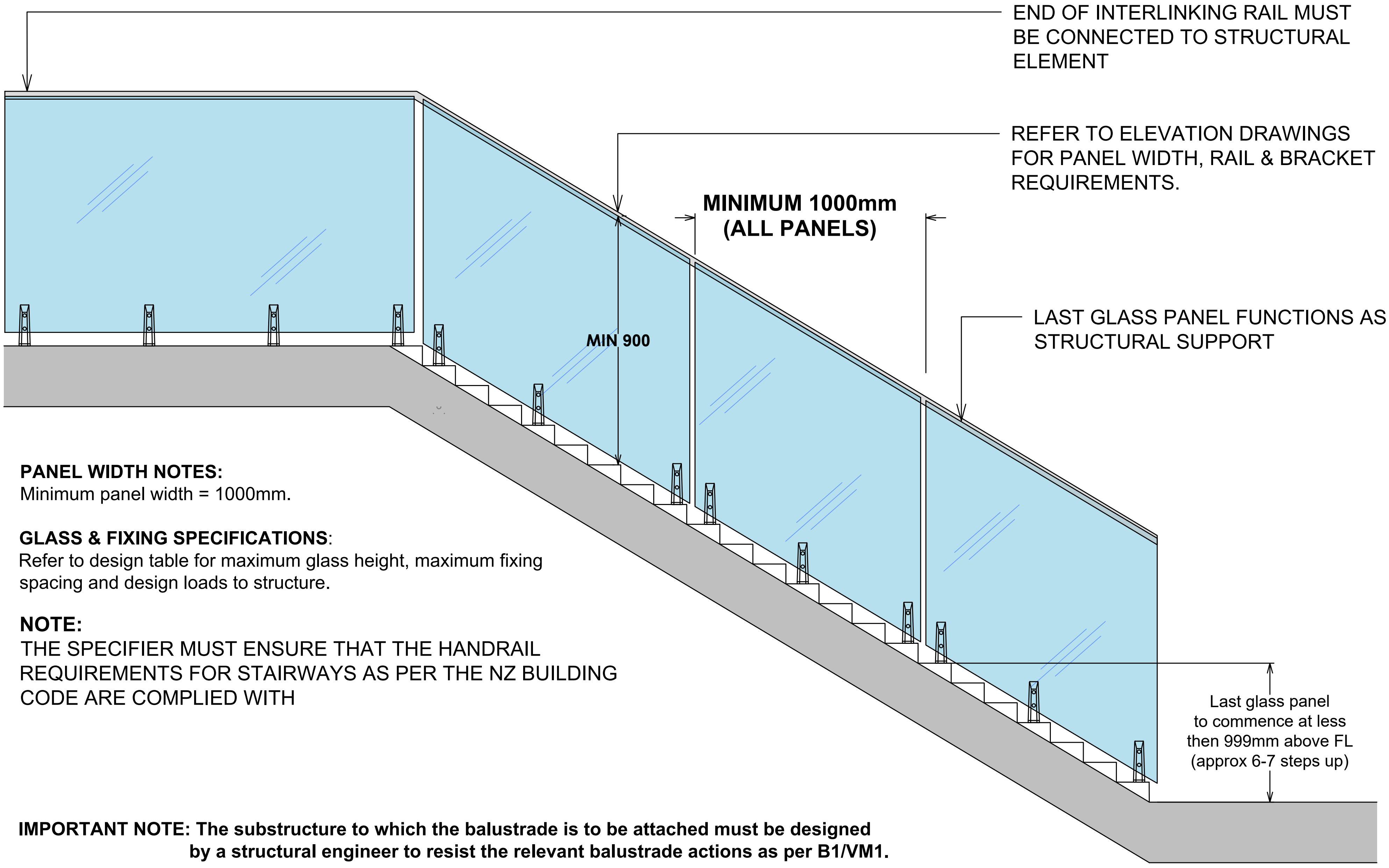
As of JAN 2017, complies with Building Code clause F9 & section 162C of the building act.



IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.

STRUT POST SYSTEM

SIDE FIXED STAIR BALUSTRADE



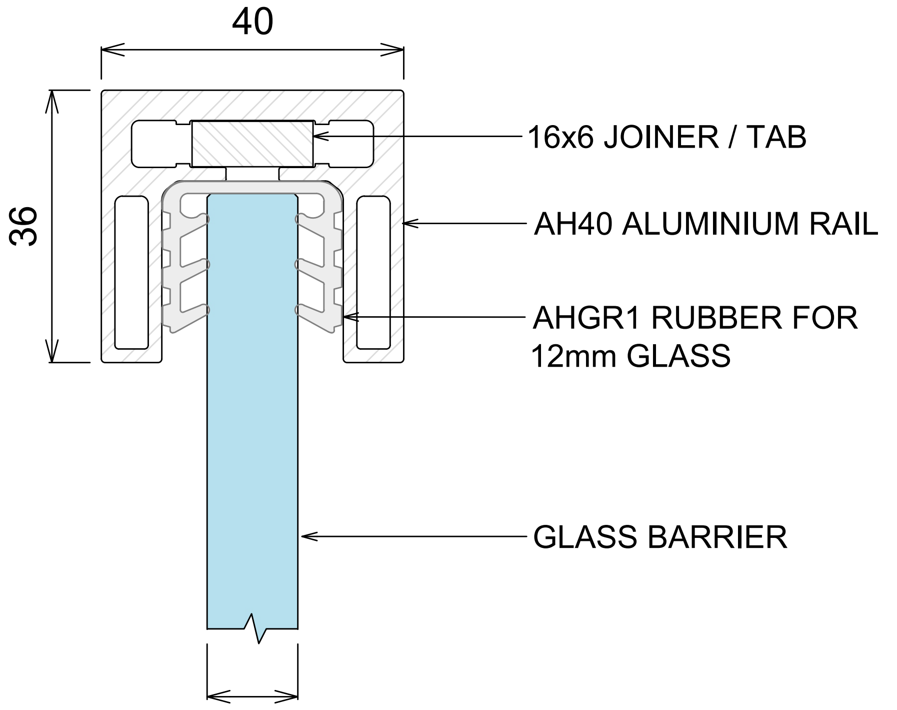
PANEL WIDTH NOTES:
Minimum panel width = 1000mm.

GLASS & FIXING SPECIFICATIONS:
Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

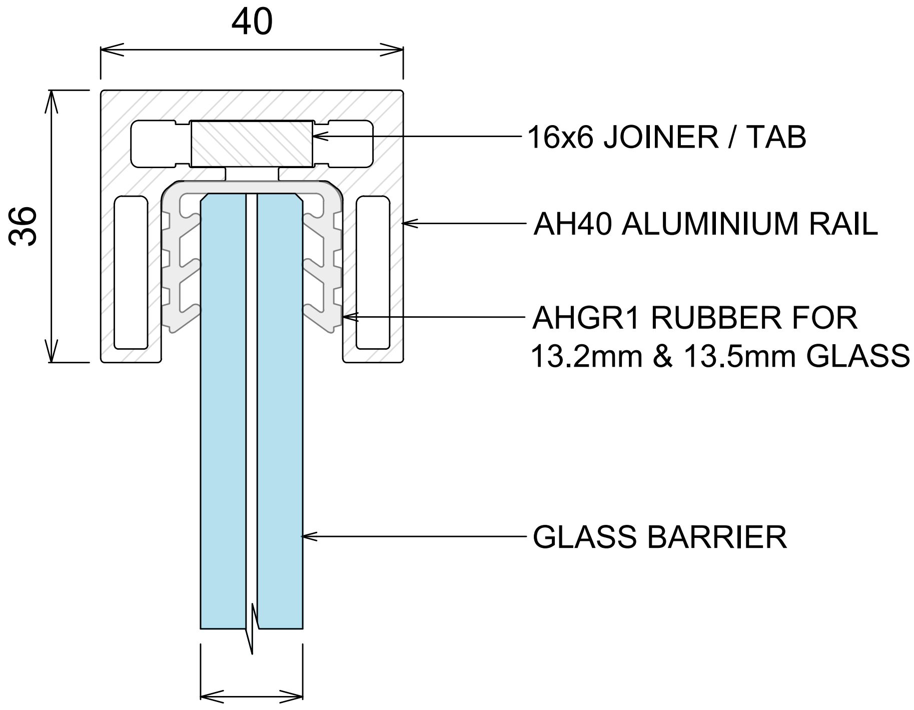
NOTE:
THE SPECIFIER MUST ENSURE THAT THE HANDRAIL REQUIREMENTS FOR STAIRWAYS AS PER THE NZ BUILDING CODE ARE COMPLIED WITH

IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.

REFER TO DESIGN TABLE & ELEVATIONS FOR LIMITATIONS & OPTIONS

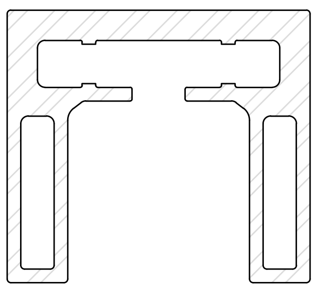


12mm Toughened Safety Glass

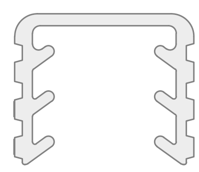


13.2mm OR 13.52mm Toughened Laminate Safety Glass

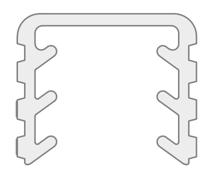
COMPONENTS



AH40 ALUMINIUM RAIL



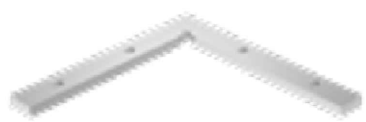
AHGR1
FOR 12mm GLASS



AHGR1
FOR 13.2mm OR 13.52mm GLASS

JOINERS ALL 16 x 6mm

AHJ180 STRAIGHT INLINE JOINER WITH LOCKING SCREWS



AHJ90 90 DEGREE JOINER WITH LOCKING SCREWS

AHAVJ ADJUSTABLE VERTICAL JOINER WITH LOCKING SCREWS

AHAHJ ADJUSTABLE HORIZONTAL JOINER WITH LOCKING SCREWS

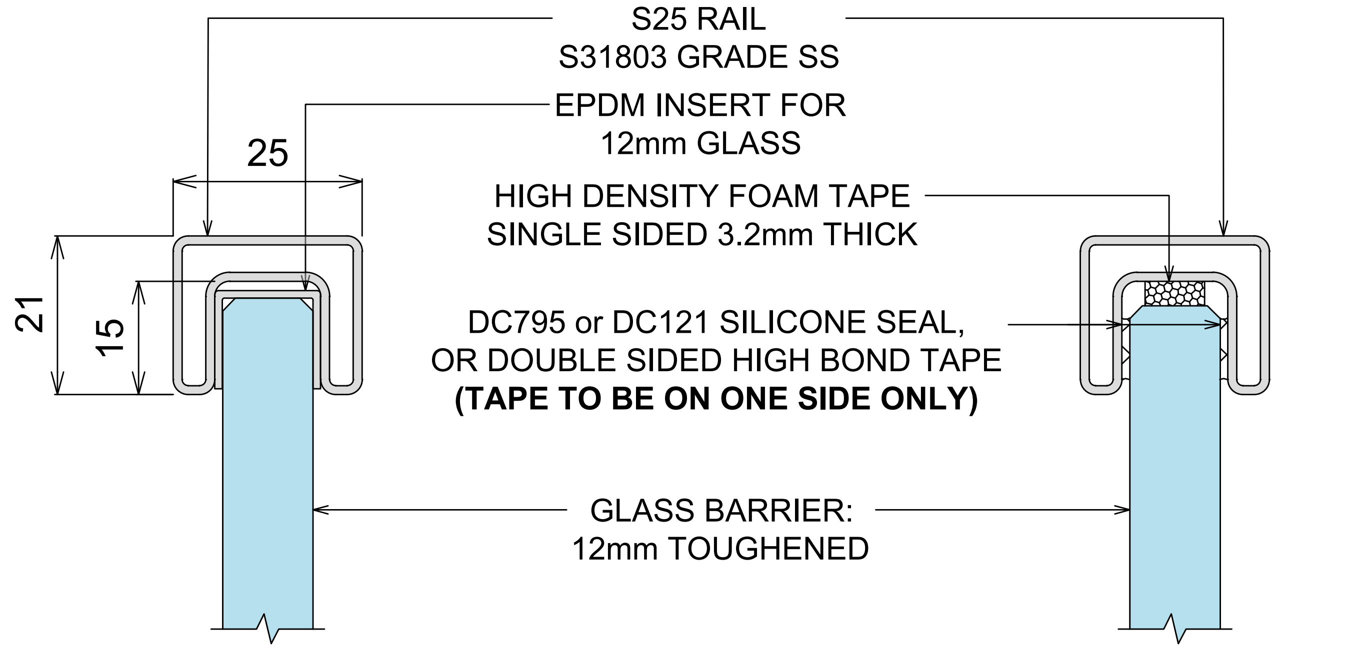
	Wall Bracket	Satin Brushed Ano	AHWB-SS
		Brushed Black Ano	AHWB-BK
	End Cap With Locking Screw	Satin Brushed Ano	AHEC-SS
		Brushed Black Ano	AHEC-BK

INSTALLATION NOTES:

1. CUT SHORT LENGTHS OF GASKET (nom 50mm) AND PLACE AT APPROXIMATELY 700mm CENTRES.
2. CUT / ADJUST INTERLINKING RAIL TO CORRECT DIMENSIONS AND TEST IN POSITION.
3. REMOVE ALL PARTS FROM GLASS BARRIER AND INSTALL FULL CUT LENGTHS OF GASKET TO TOP EDGE OF GLASS BARRIER.
4. ASSEMBLE TOP RAIL, JOINERS AND SUITABLE END PLATES.
5. PLACE BLOBS OF SEIGALSEAL N05+ V60 SILICONE IN EVERY GASKET HOLE.
6. PLACE TOP RAIL EXTRUSION, JOINERS AND END PLATES IN POSITION ON GLASS BARRIER, CLIPPING FIRMLY TO GASKET.
7. TAPE ASSEMBLED COMPONENTS DOWN TO GLASS BARRIER AND WAIT 24hrs TO FULLY BOND.
8. CLEAN UP ANY EXCESS SILICONE.

NOTE: RAIL ENDS MUST BE ATTACHED TO STRUCTURE OR STRUCTURAL POST.
EXTRUSION JOINS MUST HAVE A SUITABLE JOINER PLATE

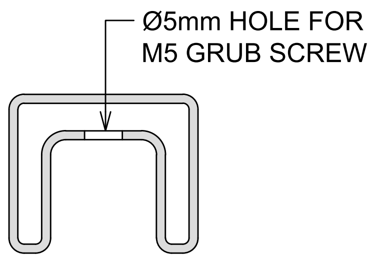
REFER TO DESIGN TABLE & ELEVATIONS FOR LIMITATIONS & OPTIONS



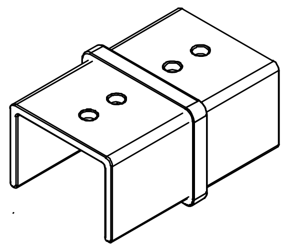
EPDM GASKET SYSTEM

TAPE & SILICONE SYSTEM

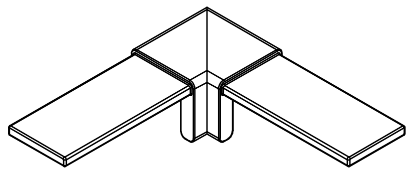
COMPONENTS OVERVIEW



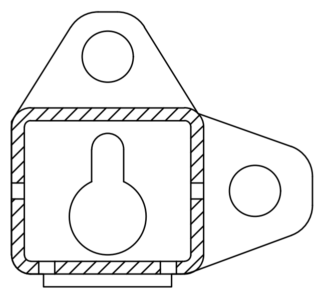
S25 ALUMINIUM RAIL



180° JOINER (& ADJUSTABLE VERTICAL JOINER)

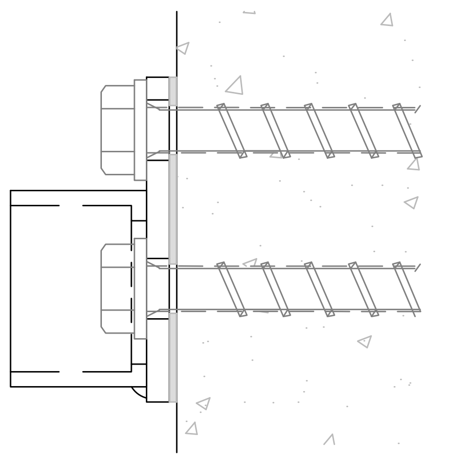


90° JOINER (& ADJUSTABLE HORIZONTAL JOINER)

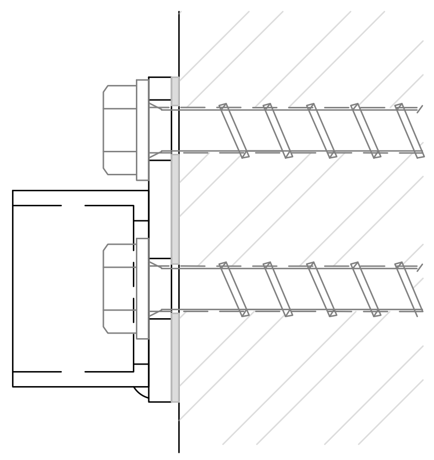


WALL BRACKET

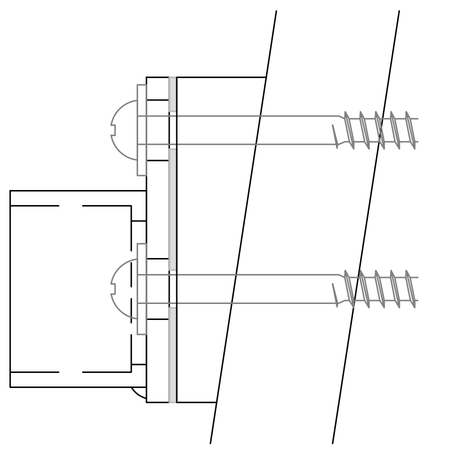
INSTALL OVERVIEW WALL BRACKET



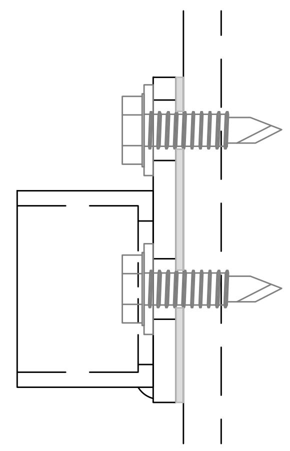
WALL BRACKET TO CONCRETE



WALL BRACKET TO BLOCKWALL



WALL BRACKET TO WEATHERBOARD



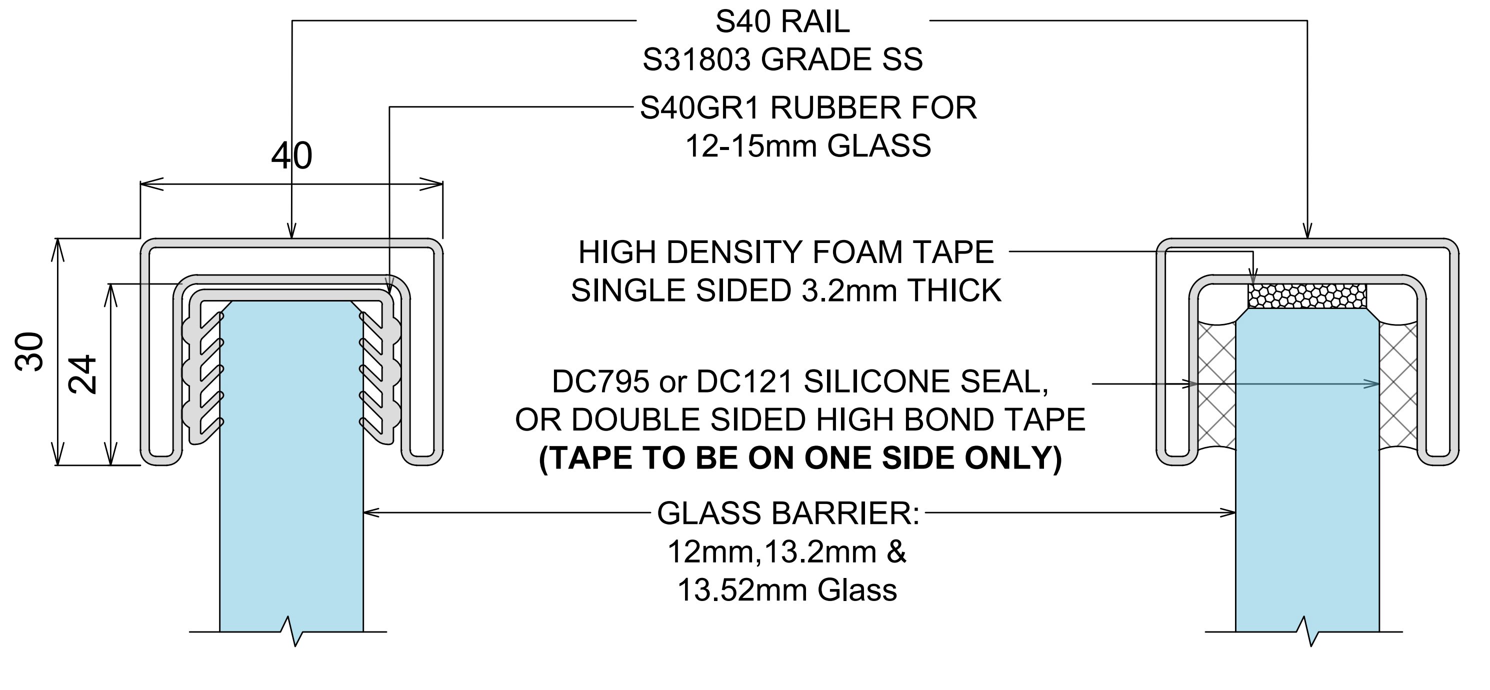
WALL BRACKET TO STEEL

NOTES:

1. PREPARE FOR AND APPLY DC795 & DC121 STRUCTURAL SILICONE IN ACCORDANCE WITH DOWSIL PREPARATION AND INSTALLATION INSTRUCTIONS.
2. INTERLINKING RAIL SPLICE & CORNER CONNECTIONS ARE SHOWN ON JOINER INSTALL OPTIONS DRAWING
3. INTERLINKING RAIL END CONNECTION BRACKETS & ATTACHMENT DETAILS ARE SHOWN ON DRAWINGS WALL BRACKET INSTALL OPTIONS. ALL SCREWS TO BE STAINLESS STEEL WITH A MINIMUM ULTIMATE SHEAR STRENGTH OF 3.5kN (PER SCREW).
4. LINK RAIL SECTION AND CONNECTION PIECES TO BE S31803 GRADE STAINLESS STEEL, IN ACCORDANCE WITH NZS 4673:2001.
5. REFER TO WARRANTY & MAINTENANCE PAGES FOR PERIODIC INSPECTION, CLEANING & MAINTENANCE REQUIREMENTS.

BALUSTRADE SYSTEM
S40 RAIL FOR 12- 21.5mm GLASS - OVERVIEW

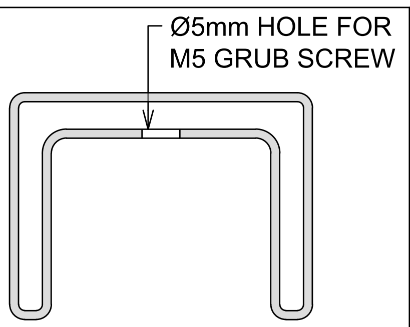
REFER TO DESIGN TABLE & ELEVATIONS FOR LIMITATIONS & OPTIONS



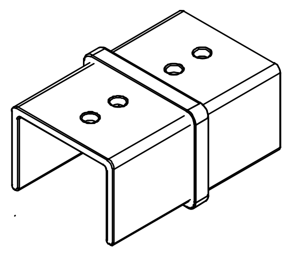
EPDM GASKET SYSTEM

TAPE & SILICONE SYSTEM

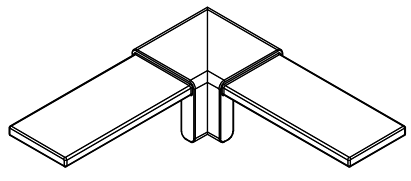
COMPONENTS OVERVIEW



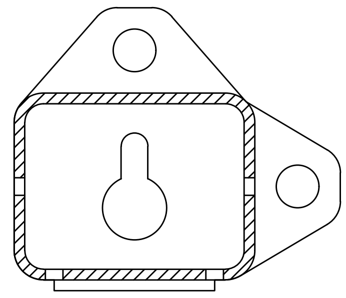
S40 ALUMINIUM RAIL



180° JOINER

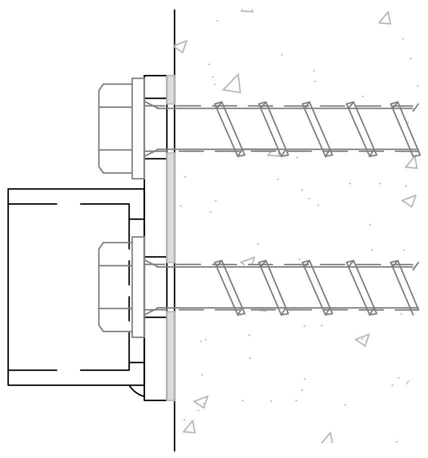


90° JOINER

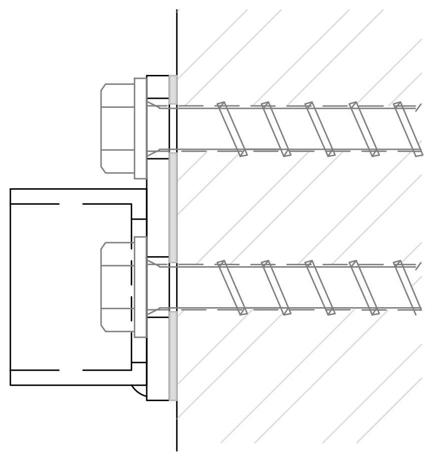


WALL BRACKET

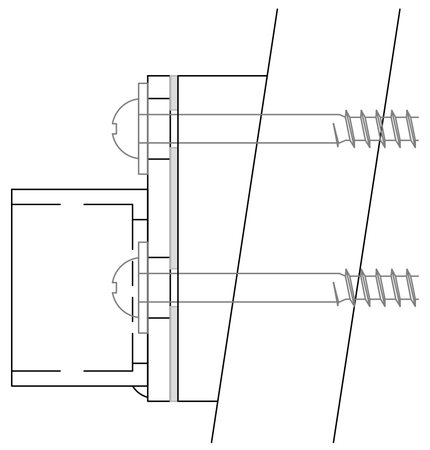
INSTALL OVERVIEW WALL BRACKET



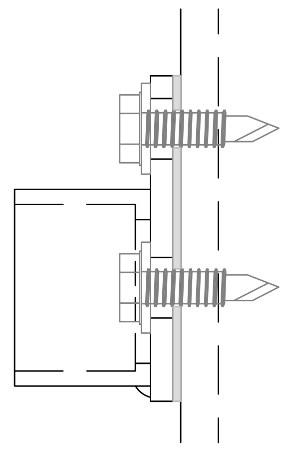
WALL BRACKET TO CONCRETE



WALL BRACKET TO BLOCKWALL



WALL BRACKET TO WEATHERBOARD



WALL BRACKET TO STEEL

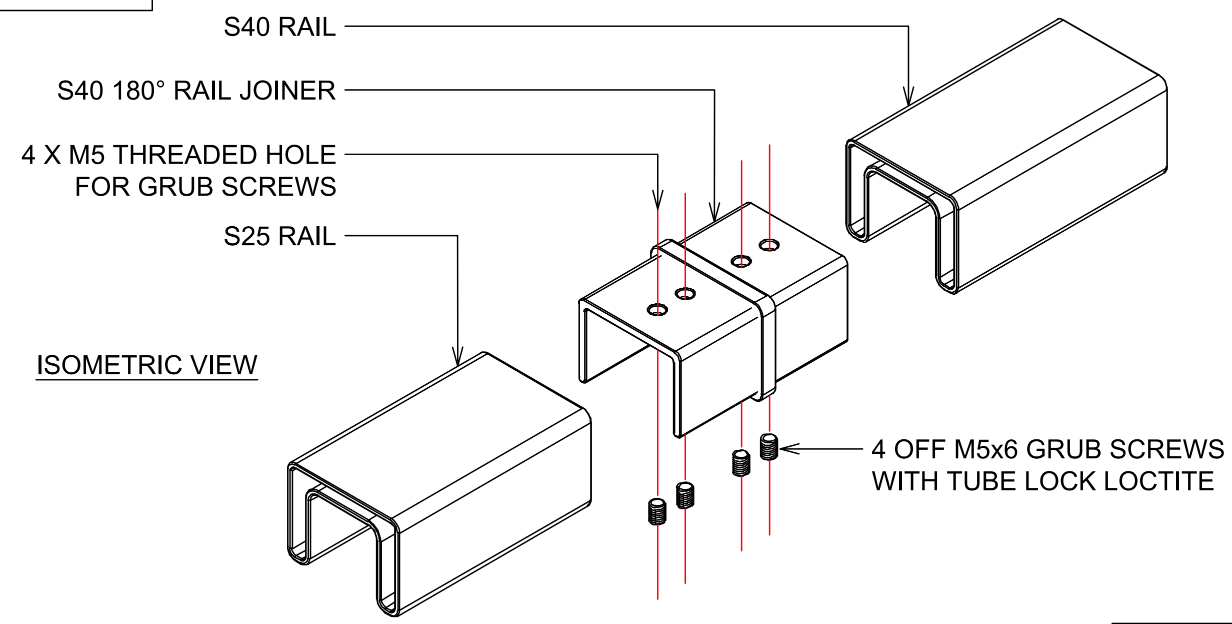
NOTES:

1. PREPARE FOR AND APPLY DC795 & DC121 STRUCTURAL SILICONE IN ACCORDANCE WITH DOWSIL PREPARATION AND INSTALLATION INSTRUCTIONS.
2. INTERLINKING RAIL SPLICE & CORNER CONNECTIONS ARE SHOWN ON JOINER INSTALL OPTIONS DRAWING
3. INTERLINKING RAIL END CONNECTION BRACKETS & ATTACHMENT DETAILS ARE SHOWN ON DRAWINGS WALL BRACKET INSTALL OPTIONS. ALL SCREWS TO BE STAINLESS STEEL WITH A MINIMUM ULTIMATE SHEAR STRENGTH OF 3.5kN (PER SCREW).
4. LINK RAIL SECTION AND CONNECTION PIECES TO BE S31803 GRADE STAINLESS STEEL, IN ACCORDANCE WITH NZS 4673:2001.
5. REFER TO WARRANTY & MAINTENANCE PAGES FOR PERIODIC INSPECTION, CLEANING & MAINTENANCE REQUIREMENTS.

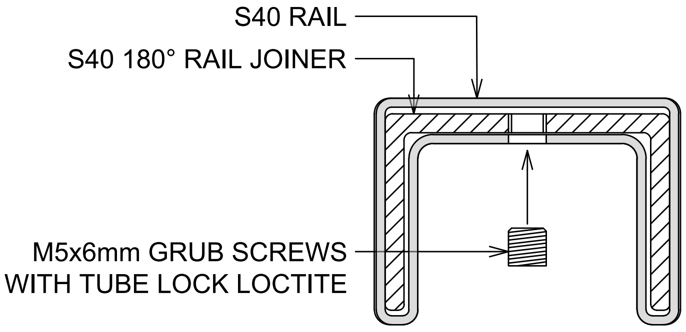
BALUSTRADE SYSTEM
S40 RAIL FOR 12- 21.5mm GLASS - JOINER DETIALS

ALL FIXINGS TO BE STAINLESS STEEL

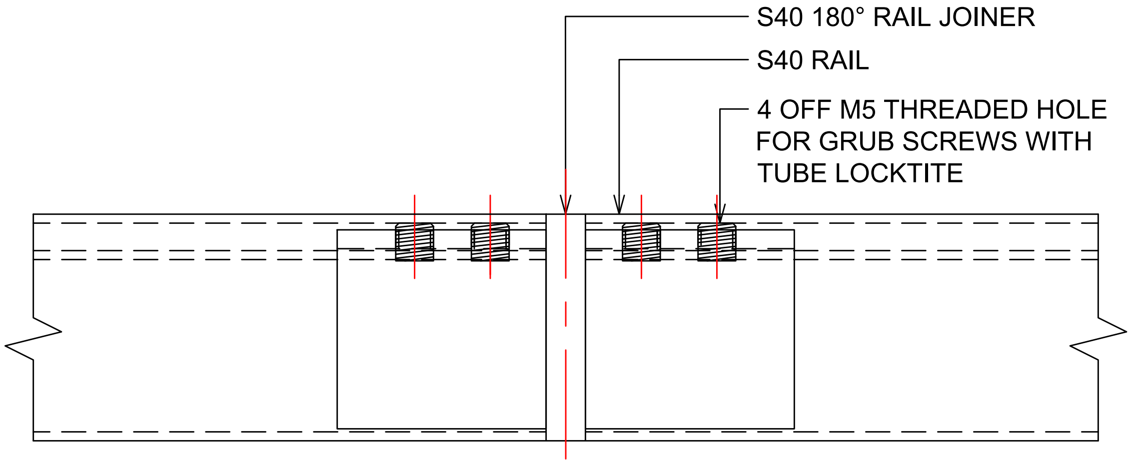
S40 180° JOINER DETAILS



ISOMETRIC VIEW



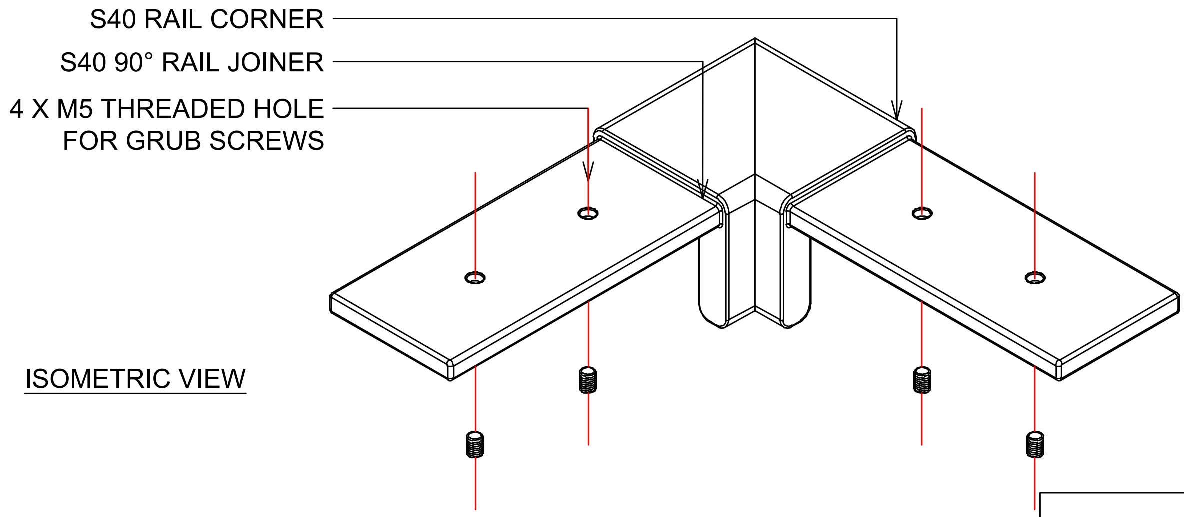
SECTION VIEW



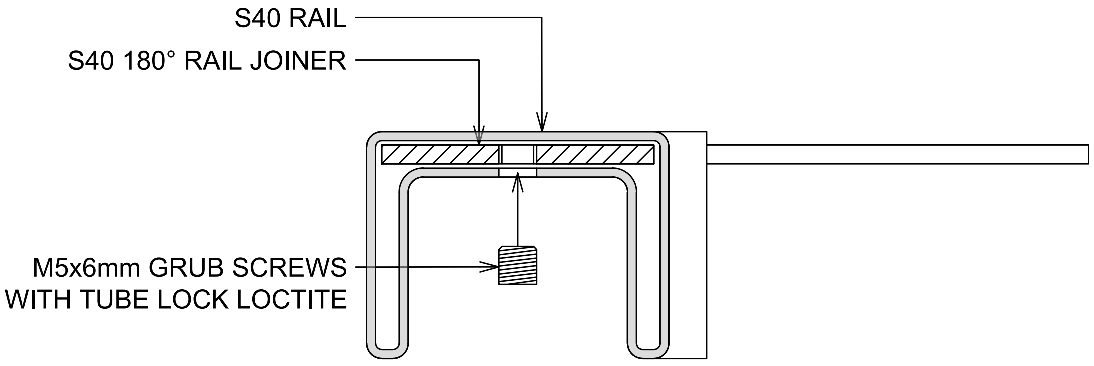
ELEVATION VIEW

ALL FIXINGS TO BE STAINLESS STEEL

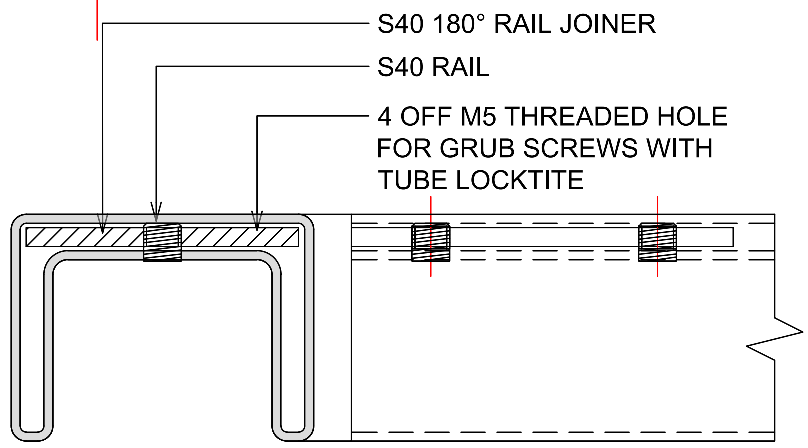
S40 90° JOINER DETAILS



ISOMETRIC VIEW



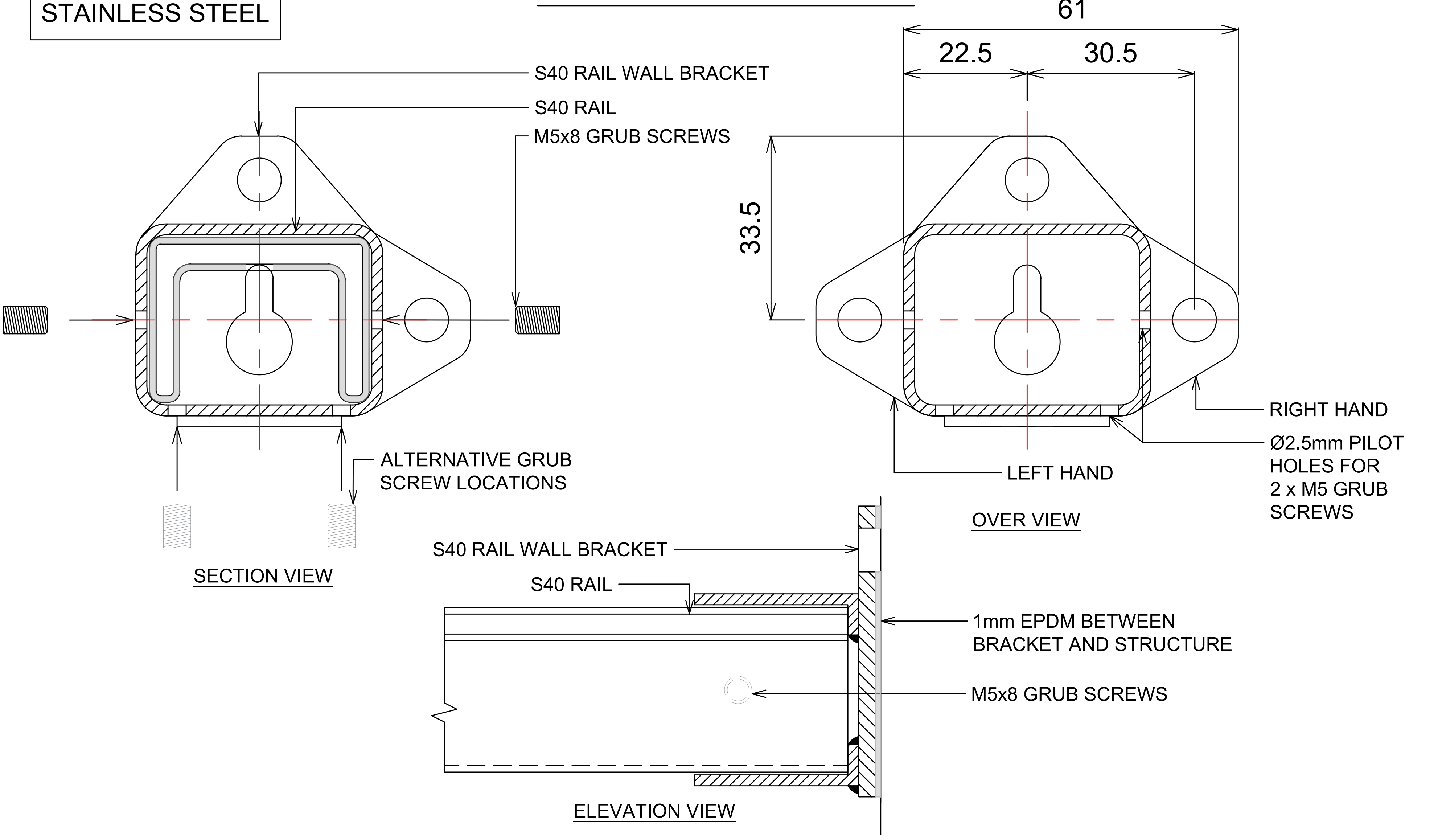
SECTION VIEW



ELEVATION VIEW

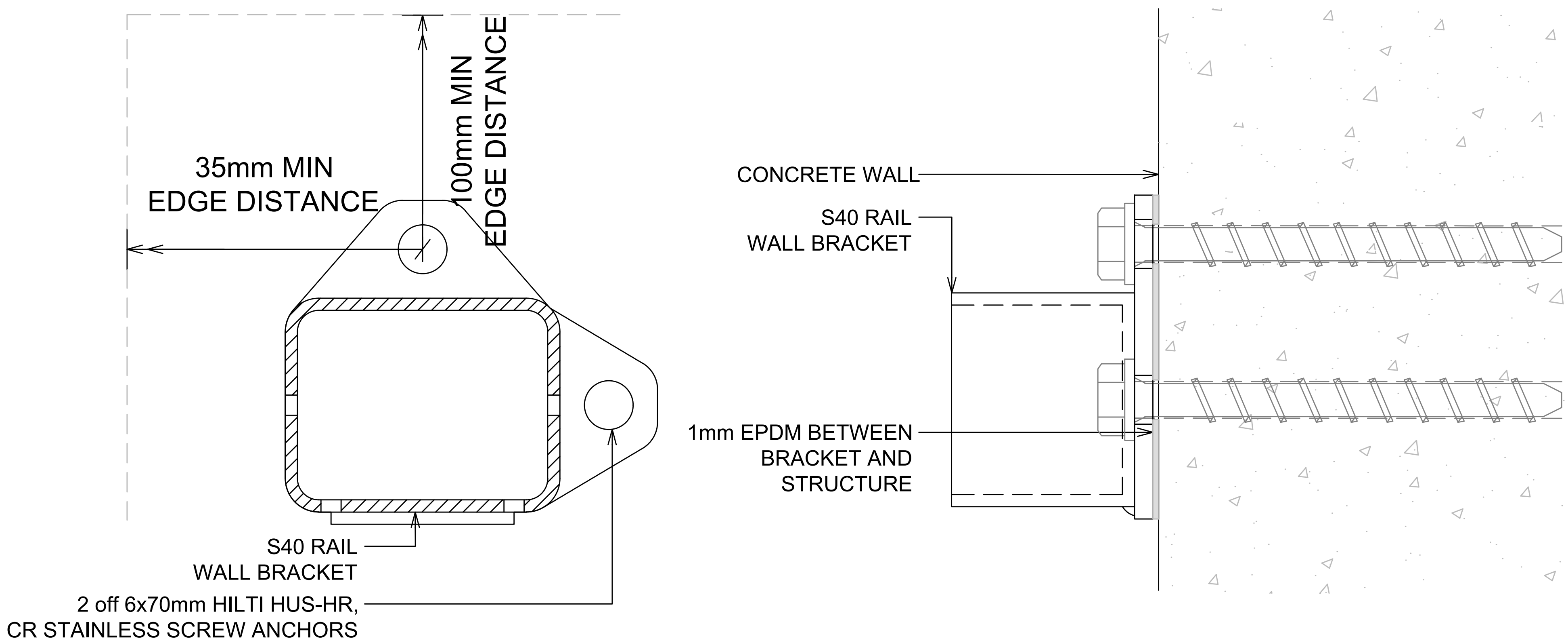
ALL FIXINGS TO BE STAINLESS STEEL

S40 RAIL WALL BRACKET



ALL FIXINGS TO BE STAINLESS STEEL

S40 RAIL WALL BRACKET TO CONCRETE WALL

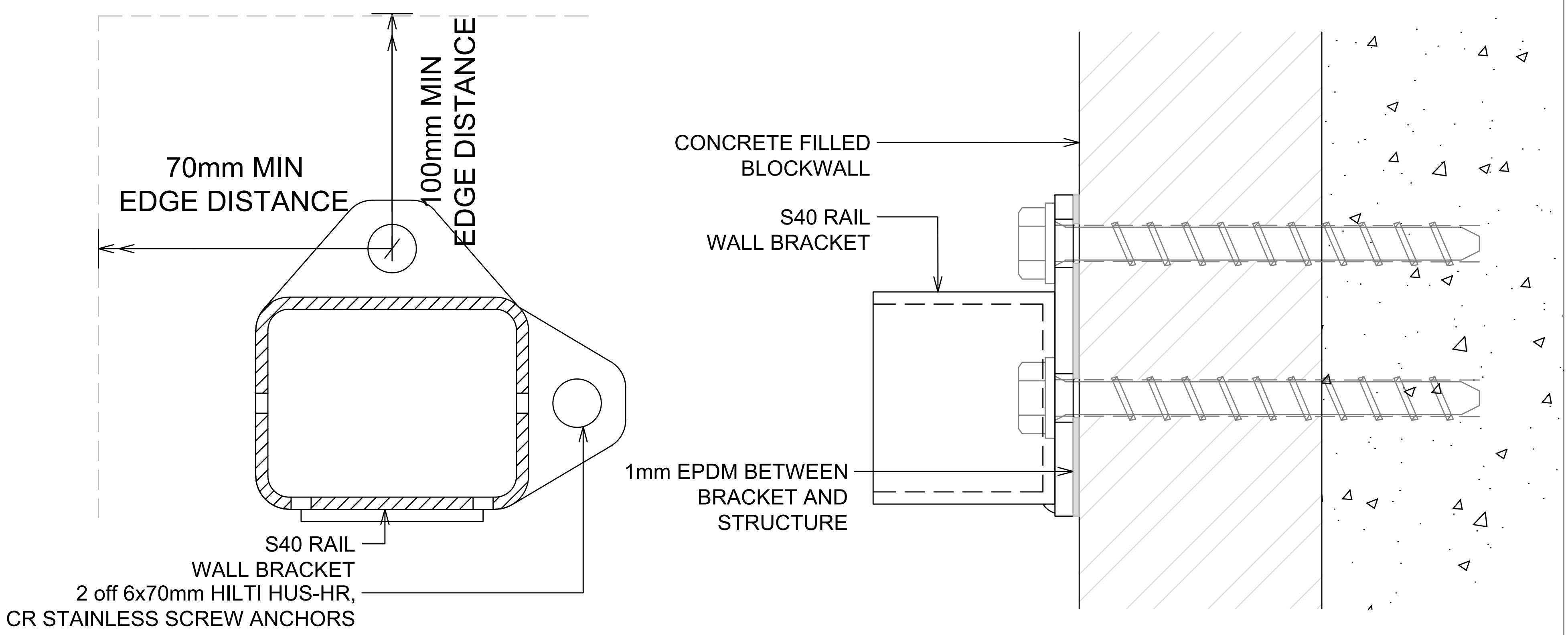


NOTES:

1. CONCRETE WALL IS TO BE DESIGNED BY PROJECT STRUCTURAL ENGINEER FOR LOADS IMPOSED BY BALUSTRADE. ULS POINT LOAD, N* = 0.9KN - INWARDS, OUTWARDS OR DOWN.
2. CONCRETE WALL TO BE MINIMUM 140mm THICK.
3. CONCRETE WALL MUST BE REINFORCED & IS TO BE DESIGNED & DETAILED IN ACCORDANCE WITH NZS3101.
4. MINIMUM CONCRETE STRENGTH = 20MPa.

ALL FIXINGS TO BE STAINLESS STEEL

S40 RAIL WALL BRACKET TO BLOCKWALL

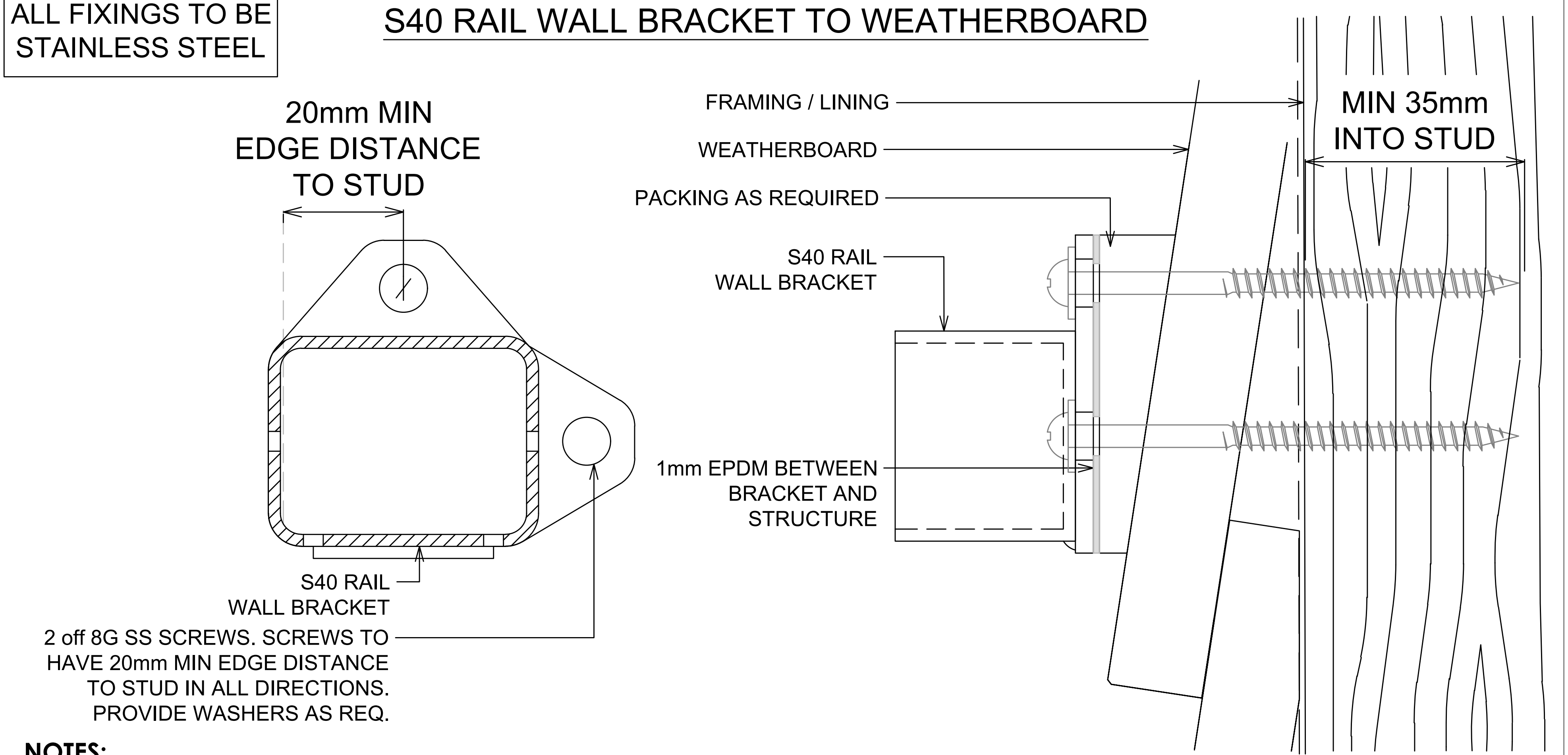


NOTES:

1. BLOCKWALL IS TO BE DESIGNED BY PROJECT STRUCTURAL ENGINEER FOR LOADS IMPOSED BY BALUSTRADE. ULS POINT LOAD, N* = 0.9KN - INWARDS, OUTWARDS OR DOWN.
2. MINIMUM BLOCKWALL THICKNESS = 140mm.
3. BLOCKWALL MUST BE COREFILLED / REINFORCED & IS TO BE DESIGNED & DETAILED IN ACCORDANCE WITH NZS4230 OR NZS4229.
4. MINIMUM COREFILL CONCRETE STRENGTH = 17.5MPa.

ALL FIXINGS TO BE STAINLESS STEEL

S40 RAIL WALL BRACKET TO WEATHERBOARD

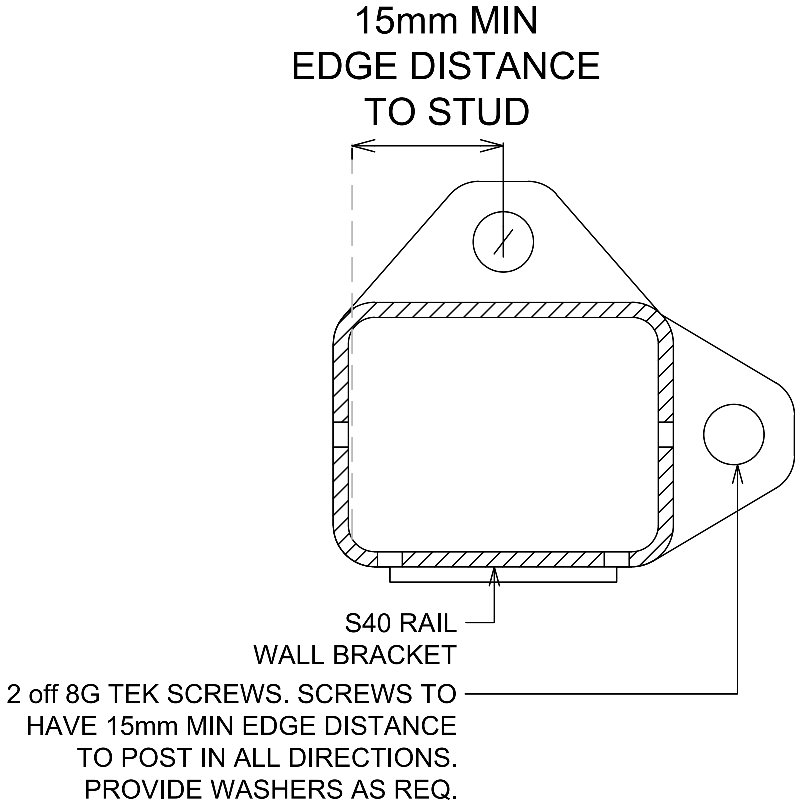


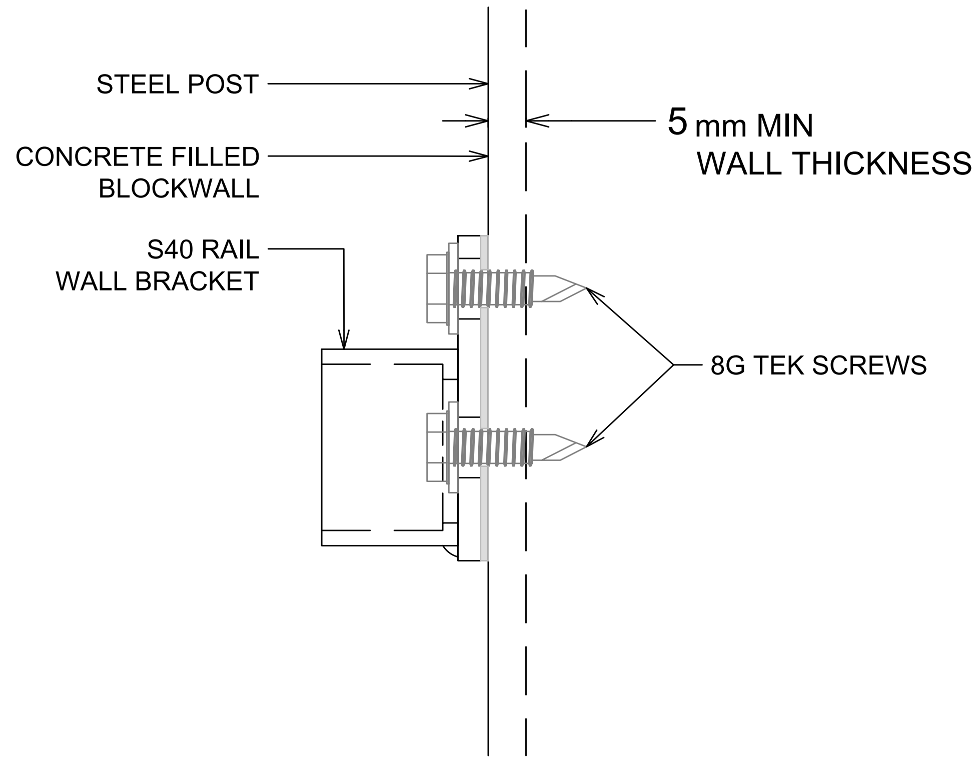
NOTES:

1. TIMBER STUD WALL IS TO BE DESIGNED BY PROJECT STRUCTURAL ENGINEER FOR LOADS IMPOSED BY BALUSTRADE. ULS POINT LOAD, N* = 0.9KN - INWARDS, OUTWARDS OR DOWN.
2. MINIMUM STUD SIZE = 90x45.
3. MINIMUM TIMBER GRADE = SG8.
4. TIMBER STUD WALL TO BE DESIGNED & DETAILED IN ACCORDANCE WITH NZS3603 OR NZS3604.

ALL FIXINGS TO BE STAINLESS STEEL

S40 RAIL WALL BRACKET TO BLOCKWALL

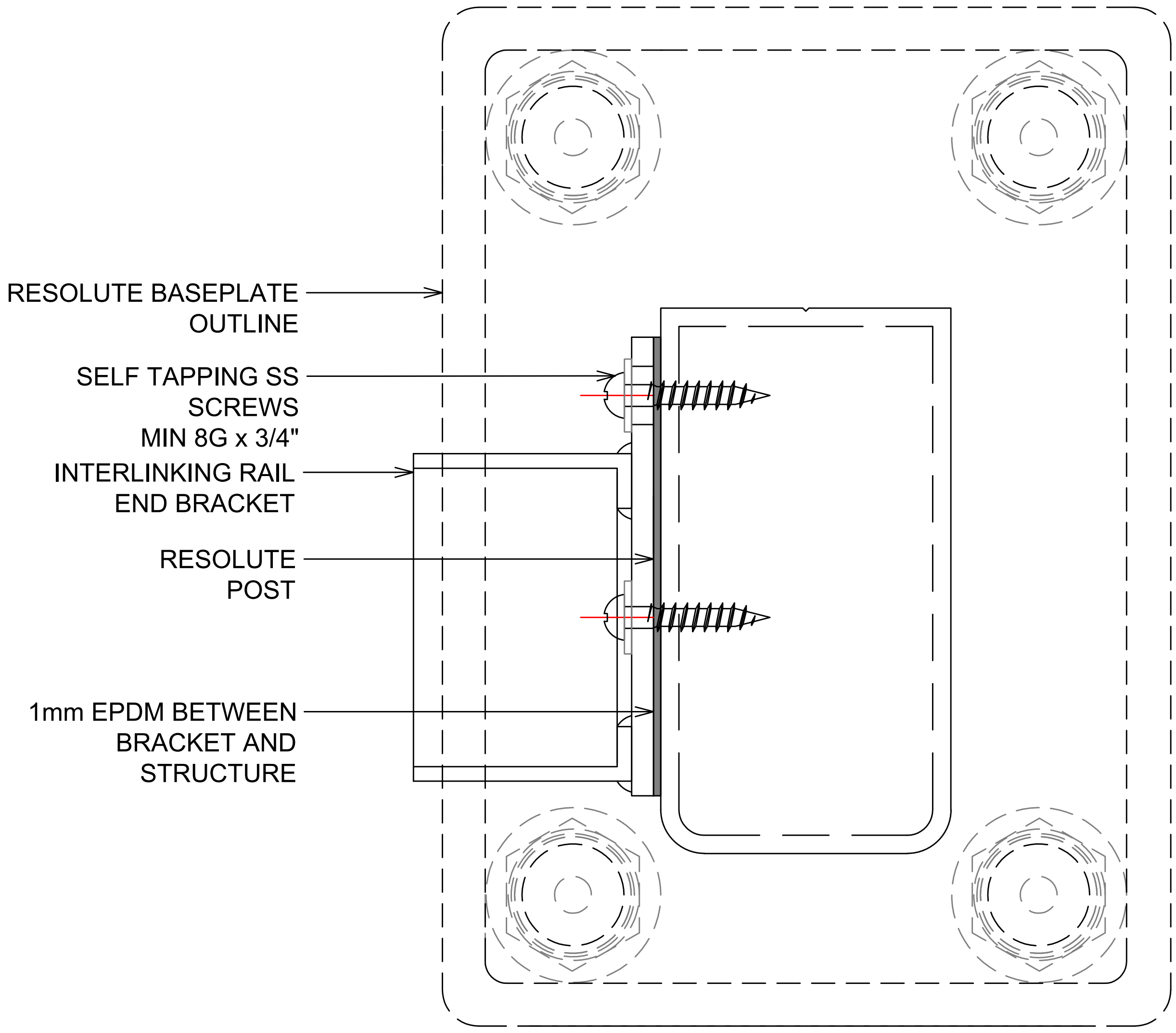




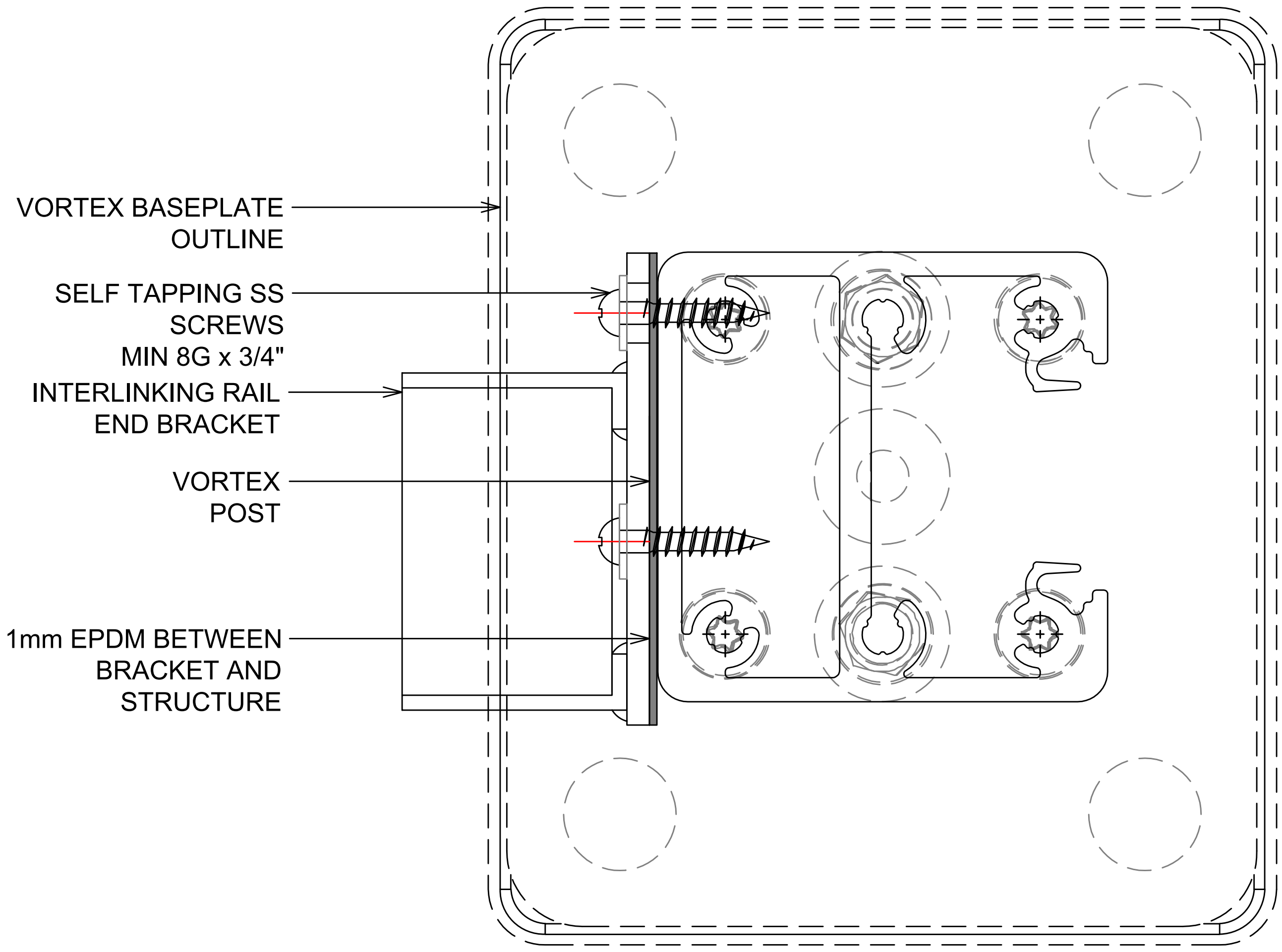
NOTES:

1. STEEL POST IS TO BE DESIGNED BY PROJECT STRUCTURAL ENGINEER FOR LOADS IMPOSED BY BALUSTRADE. ULS POINT LOAD, N* = 0.9KN - INWARDS, OUTWARDS OR DOWN.
2. BUILDING DESIGNER TO ENSURE DURABILITY REQUIREMENTS OF CONNECTION ARE MET.
3. MINIMUM STEEL POST WALL THICKNESS = 5mm.
4. MINIMUM STEEL GRADE = 300MPa.

S40 RAIL WALL BRACKET TO RESOLUTE POST SYSTEM



S40 RAIL WALL BRACKET TO VORTEX POST SYSTEM

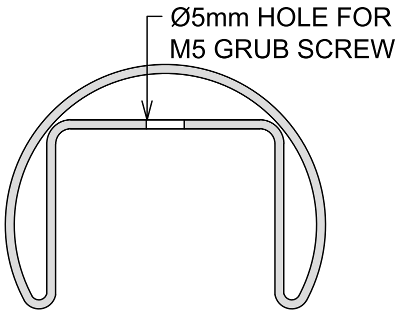


REFER TO DESIGN TABLE & ELEVATIONS FOR LIMITATIONS & OPTIONS

EPDM GASKET SYSTEM

TAPE & SILICONE SYSTEM

COMPONENTS OVERVIEW



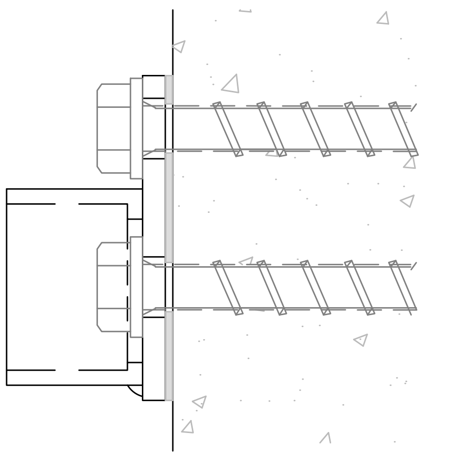
S40 ALUMINIUM RAIL

180° JOINER

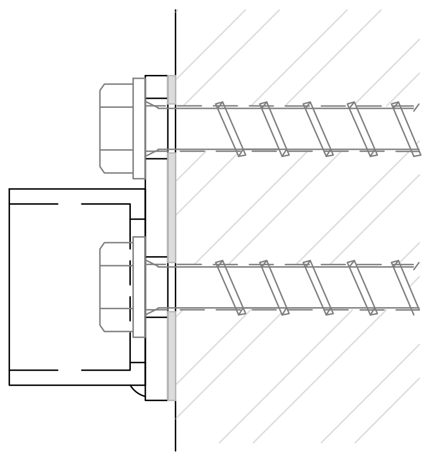
90° JOINER

WALL BRACKET

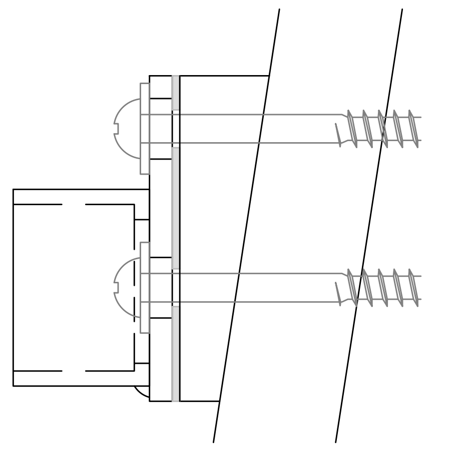
INSTALL OVERVIEW WALL BRACKET



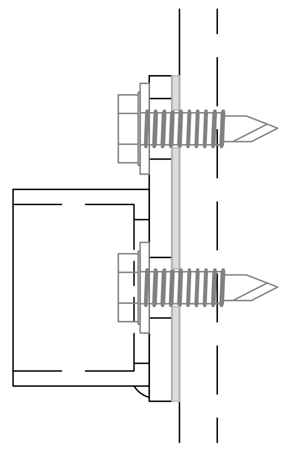
WALL BRACKET TO CONCRETE



WALL BRACKET TO BLOCKWALL



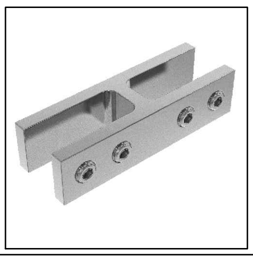
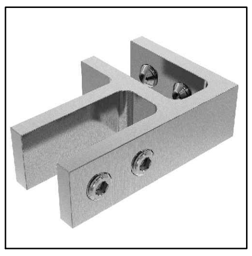
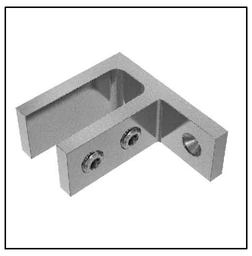
WALL BRACKET TO WEATHERBOARD



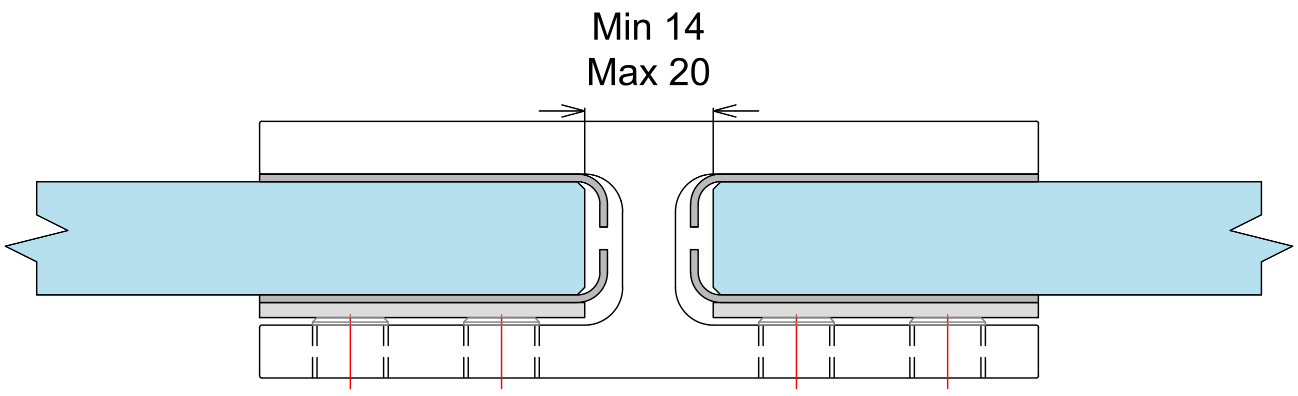
WALL BRACKET TO STEEL

NOTES:

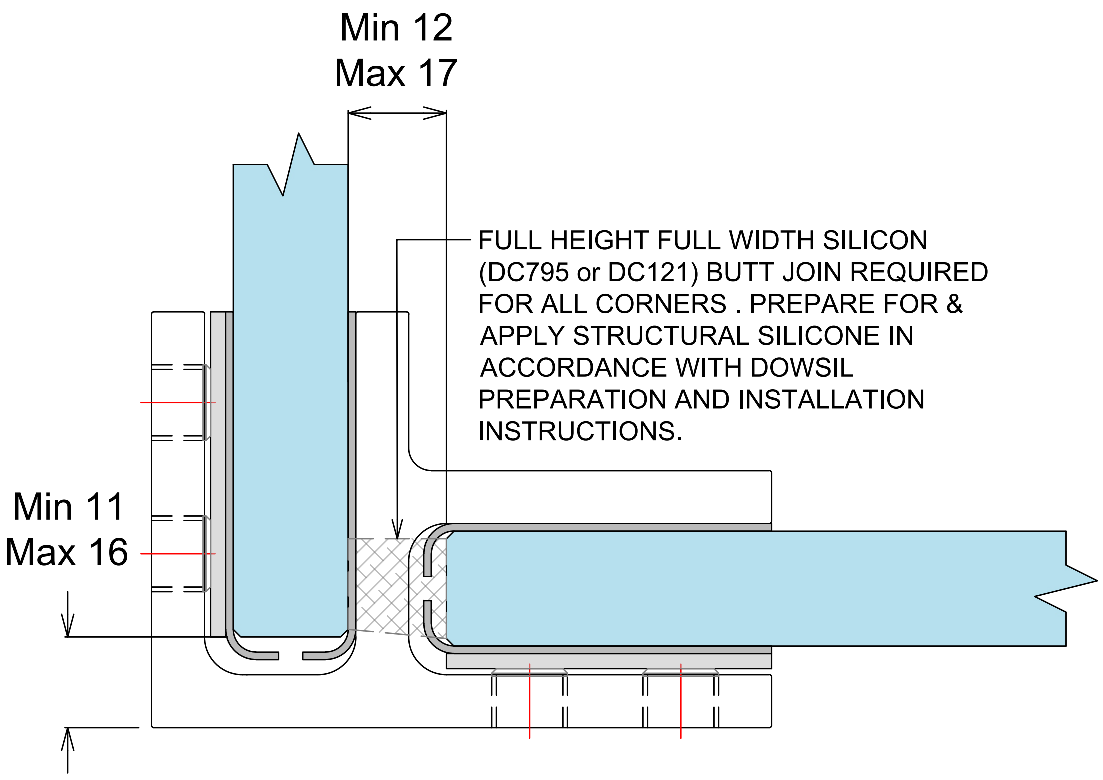
1. PREPARE FOR AND APPLY DC795 & DC121 STRUCTURAL SILICONE IN ACCORDANCE WITH DOWSIL PREPARATION AND INSTALLATION INSTRUCTIONS.
2. INTERLINKING RAIL SPLICE & CORNER CONNECTIONS ARE SHOWN ON JOINER INSTALL OPTIONS DRAWING
3. INTERLINKING RAIL END CONNECTION BRACKETS & ATTACHMENT DETAILS ARE SHOWN ON DRAWINGS WALL BRACKET INSTALL OPTIONS. ALL SCREWS TO BE STAINLESS STEEL WITH A MINIMUM ULTIMATE SHEAR STRENGTH OF 3.5kN (PER SCREW).
4. LINK RAIL SECTION AND CONNECTION PIECES TO BE S31803 GRADE STAINLESS STEEL, IN ACCORDANCE WITH NZS 4673:2001.
5. REFER TO WARRANTY & MAINTENANCE PAGES FOR PERIODIC INSPECTION, CLEANING & MAINTENANCE REQUIREMENTS.

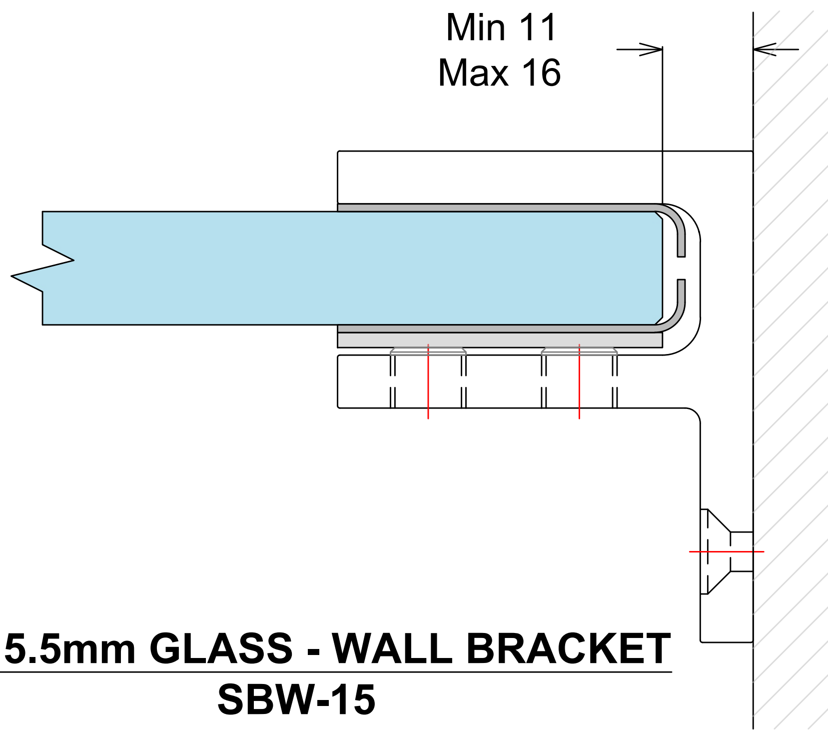
Product	Code	Description
	SB180-15 (180° bracket)	Suit glass from 12 - 15.5mm. Stiffener bracket - No holes required. 180° glass to glass. Duplex 2205. Stainless Steel
	SB90-15 (90° bracket)	Suit glass from 12 - 15.5mm. Stiffener bracket - No holes required. 90° glass to glass. Duplex 2205. Stainless Steel
	SBW-15 (wall bracket)	Suit glass from 12 - 15.5mm. Stiffener bracket - No holes required. 90° glass to wall. Duplex 2205. Stainless Steel

All brackets are supplied with a selection of gaskets to suit glass thickness and includes stainless steel clamping plates.



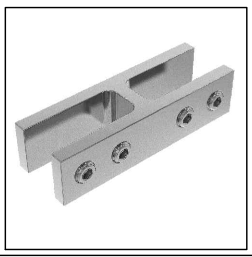
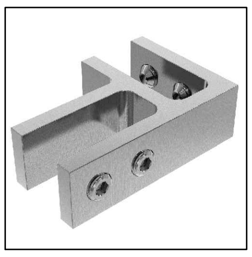
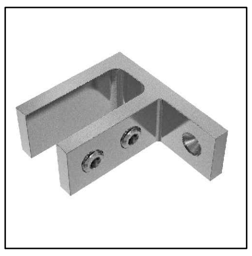
12-15.5mm GLASS - STRAIGHT BRACKET
SB180-15



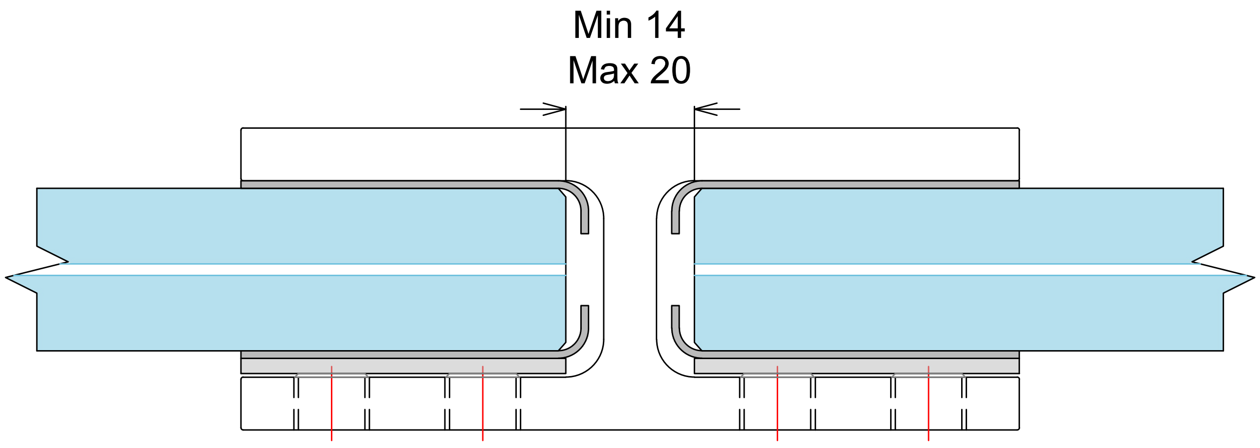


12-15.5mm GLASS - CORNER BRACKET
SB90-15

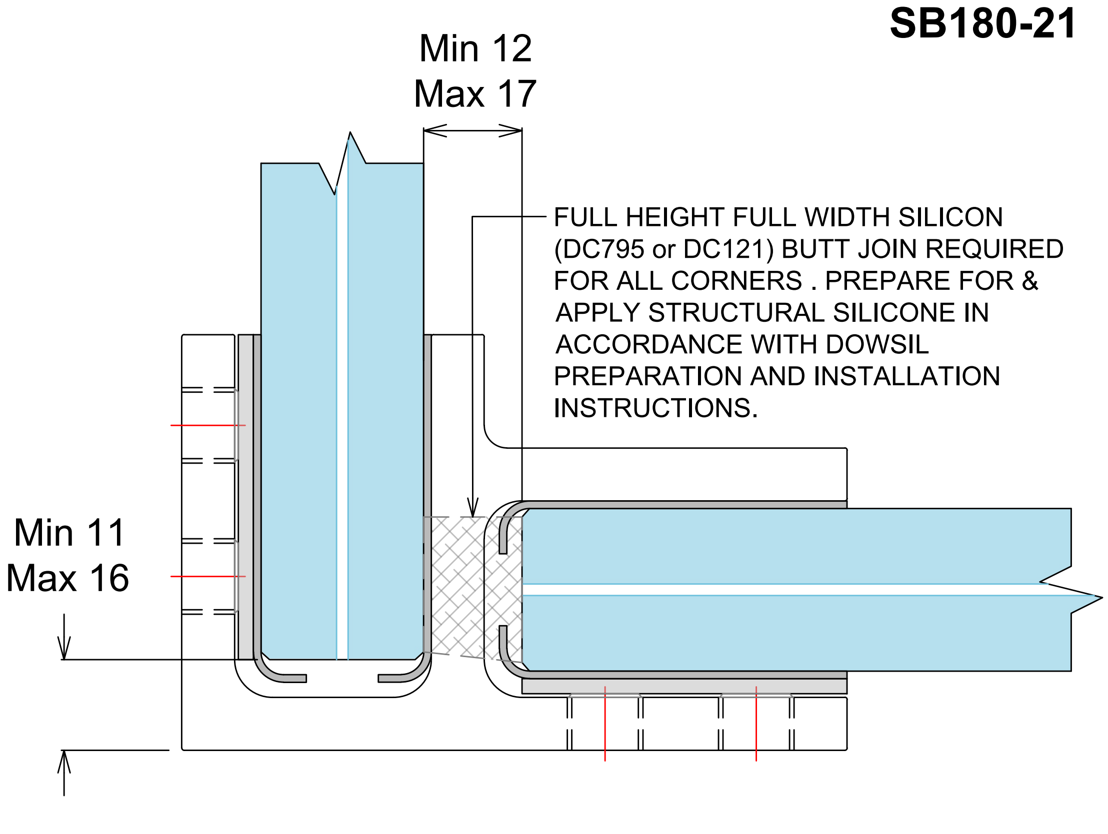
12-15.5mm GLASS - WALL BRACKET
SBW-15

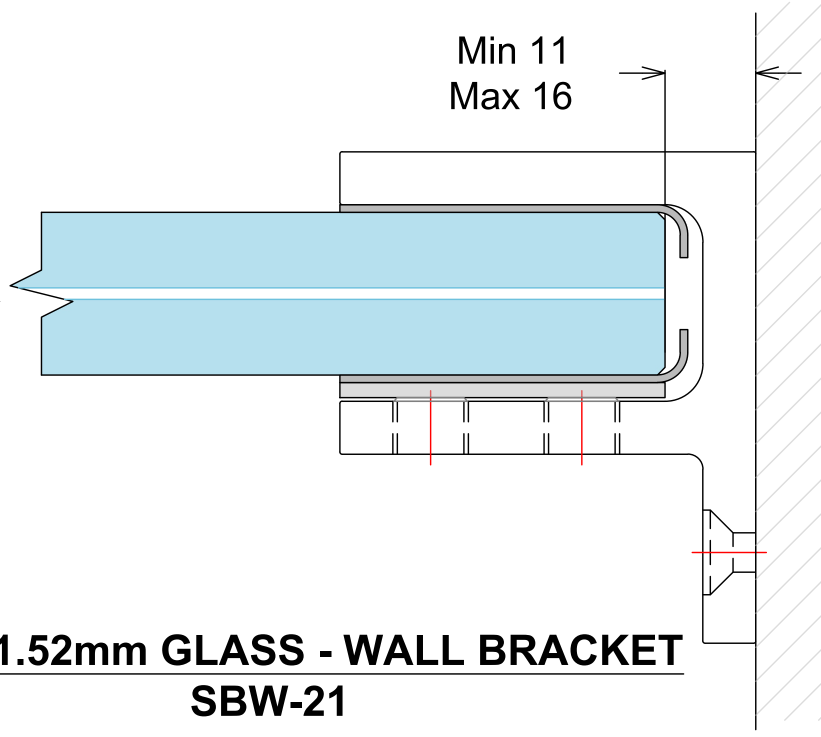
Product	Code	Description
	SB180-21 (180° bracket)	Suit glass from 17 - 22mm. Stiffener bracket - No holes required. 180° glass to glass. Duplex 2205. Stainless Steel
	SB90-21 (90° bracket)	Suit glass from 17 - 22mm. Stiffener bracket - No holes required. 90° glass to glass. Duplex 2205. Stainless Steel
	SBW-21 (wall bracket)	Suit glass from 17 - 22mm. Stiffener bracket - No holes required. 90° glass to wall. Duplex 2205. Stainless Steel

All brackets are supplied with a selection of gaskets to suit glass thickness and includes stainless steel clamping plates.



17.2 - 21.52mm GLASS - STRAIGHT BRACKET
SB180-21

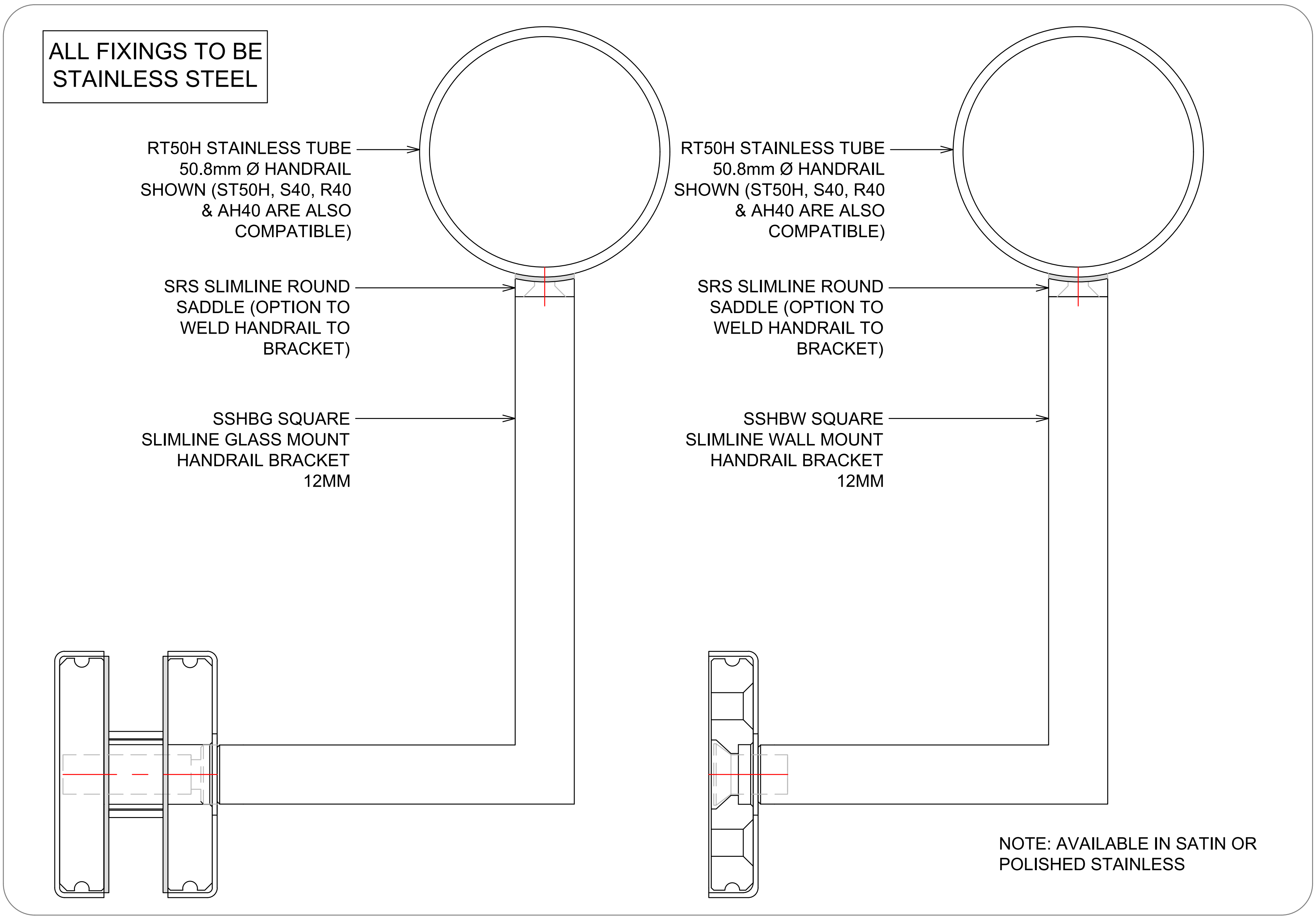


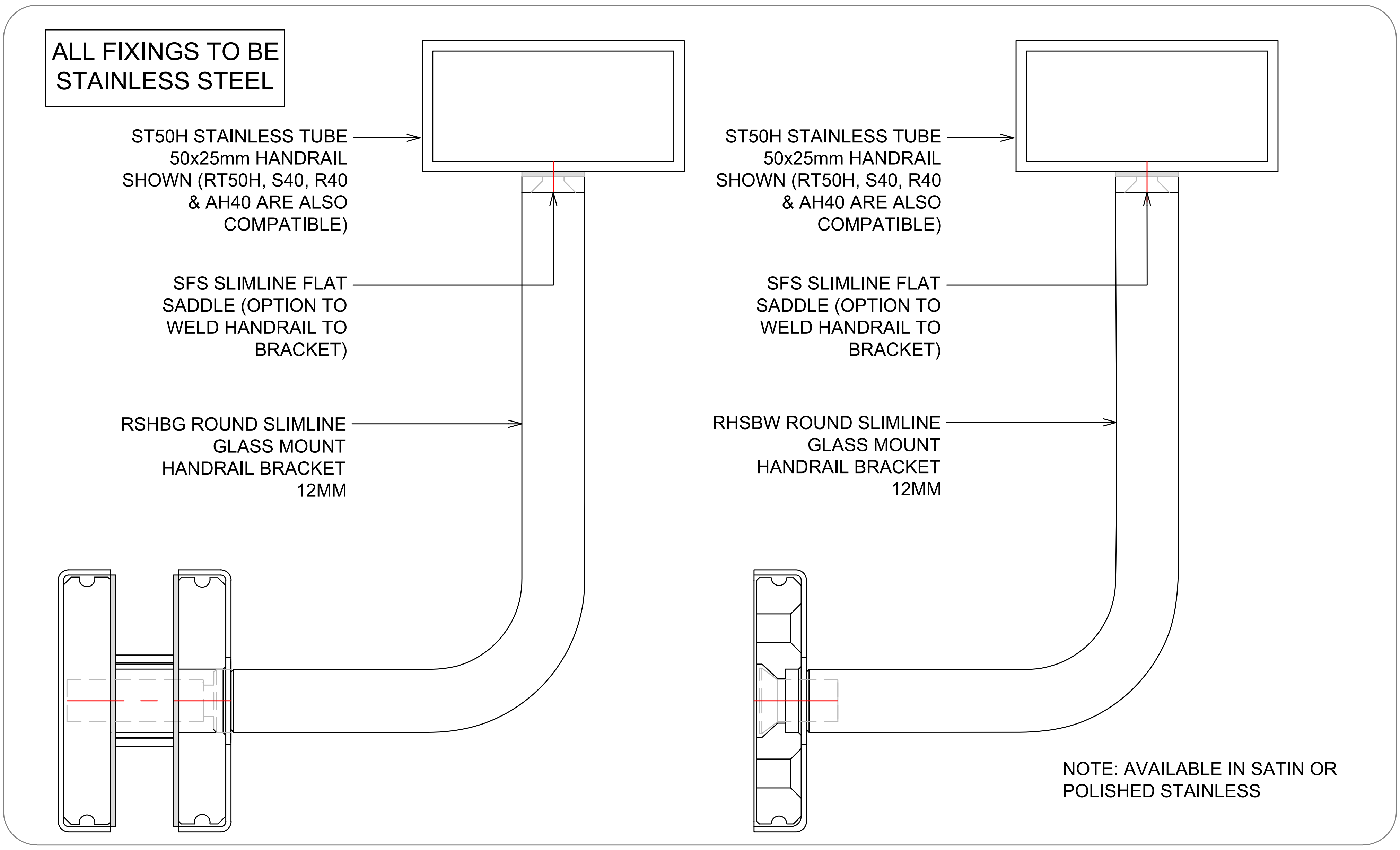


17.2 - 21.52mm GLASS - CORNER BRACKET
SC90-21

17.2 - 21.52mm GLASS - WALL BRACKET
SBW-21

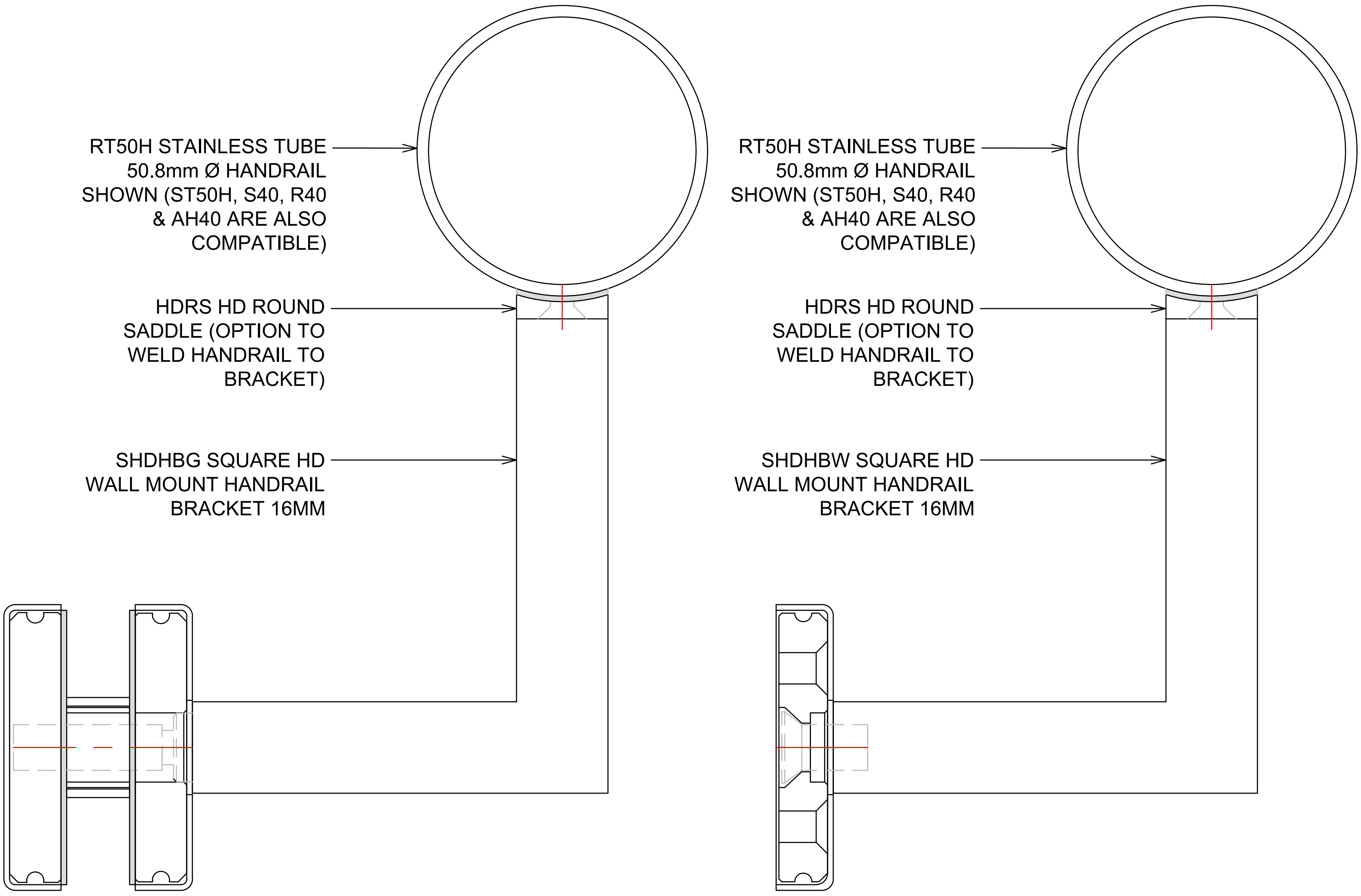
BALUSTRADE SYSTEM
SLIMLINE HANDRAIL BRACKETS



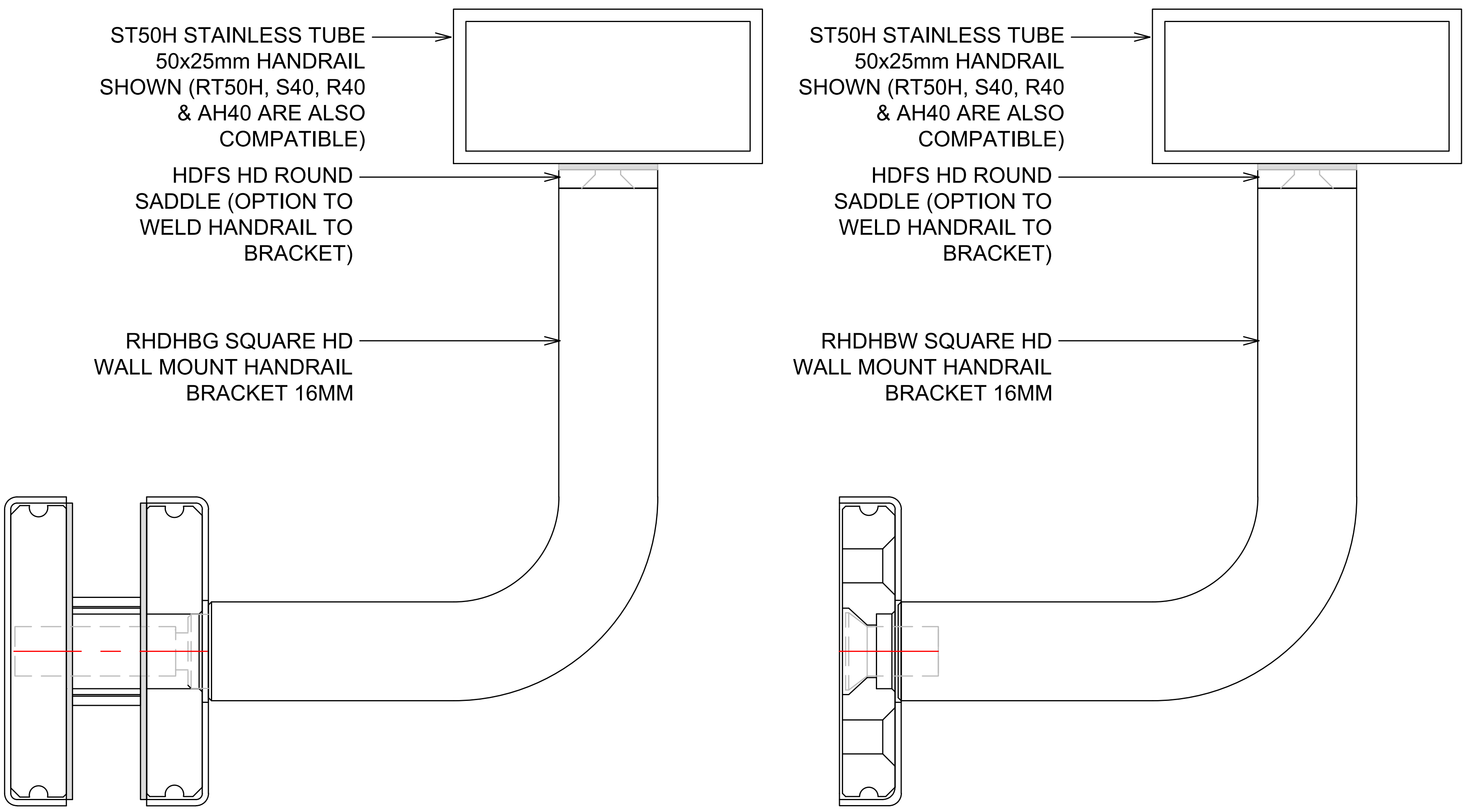


BALUSTRADE SYSTEM
HD HANDRAIL BRACKETS

ALL FIXINGS TO BE STAINLESS STEEL



ALL FIXINGS TO BE STAINLESS STEEL



CARE AND MAINTENANCE OF BALUSTRADE GLASS & HARDWARE

Hardware requires regular maintenance to ensure the system performs at its best. As a general rule, the harsher the environment, the more regular the maintenance required to keep your hardware in top condition. Also hardware or systems that are covered by verandas or wide eaves and not subject to natural rain wash needs regular cleaning to avoid damage to surface finish on both the aluminium and any surface coated hardware.

Fair wear and tear;

- **Seals and rubbers** will require replacing from time to time depending on the environment. As a general rule, they should last for 10 years or more, and can be replaced by service provider.
- **Tracks, rollers** (if accessible) and hardware require lubrication; rollers may require replacing due to normal wear and tear. This depends on the environment and amount of use.

CARE AND MAINTENANCE, WASHING GLASS

Regular washing and drying of glass windows and doors are required to ensure their long term durability. In urban areas washing should be done every three to six months. The following guidelines apply:

a. When washing, soak the glass surface with warm water and a mild soap detergent solution or proprietary glass cleaners to loosen dirt and debris.

b. Use a soft grit free cloth or sponge when washing and try to avoid washing In direct sunlight. Do not use scrapers or razor blades.

c. After washing, rinse with clean water and then dry the glass using a clean, grit-free squeegee, cloth, or paper towel. Remember, wet glass is dirty glass.

d. All water and cleaning solution residue should be dried from the window gaskets, sealants, and frames to prevent water spots.

e. Avoid cleaning tinted and reflective glass surfaces in direct sunlight. When washing special glass, the following guidelines apply:

f. When washing double glazing and laminated glass, use the same procedures as above but ensure no solvents come into contact with the edge laminate interlayer or unit sealant.

g. With reflective or Low E coated surfaces, exercise special care when cleaning - special cleaners may be required as they can be hard to clean. Follow manufacturer's instructions.

h. It is advisable to check that frame drainage is not blocked this can affect laminate and insulated glass units.

DO NOT:

- Do Not Use Scrapers of any type or size on a Glass surface.
- Do Not - Leave building dirt or residues to remain on Glass for a period of time.
- Do Not - Begin cleaning glass until you have identified the surface type.
- Do Not - Clean Glass surfaces in direct sunlight.
- Do Not - Allow dirty water or cleaning residues to remain on the Glass.
- Do Not - Begin cleaning before rinsing off loose residues.
- Do Not - Use abrasive cleaning solutions, materials or solvents.
- Do Not - Allow metal parts of the cleaning equipment to come in contact with the Glass.
- Do Not - Trap abrasive particles between the cleaning material and the Glass.

DO:

- Clean glass promptly when dirt or building residues appear.
- Determine glass surface type.
- Exercise special care when cleaning coated surfaces.
- Avoid cleaning glass surfaces in direct sunlight.
- Start cleaning at the top of a building, then continue to lower levels.
- Soak the glass surface in a clean soapy solution before cleaning.
- Use a mild non abrasive commercial cleaner.
- Use a squeege to remove all cleaning solution.
- Try your procedures on a small window and check.

CARE AND MAINTENANCE OF POWDERCOATING

Powder coating is available in a wide range of colours with commercially available surface integrity warranties from 10 to 30 years. The powder coating surface warranty period is conditional upon the formulation and micron thickness. Over time with exposure to the elements, powder coatings may show signs of weathering such as loss of gloss, chalking and slight color change. A simple regular clean will minimise the effects of weathering and will remove dirt, grime and other build-up detrimental to all powder coatings.

The frequency of such cleaning will depend on many factors including the:

- Geographical location of the building.
- Environment surrounding the building e.g. marine, industrial, alkaline or acidic, etc.
- Levels of atmospheric pollution including salts.
- Prevailing winds and the possibility of air borne debris causing erosive wear of the coating e.g. sand causing abrasion.
- Protection of part or all of the building by other buildings.
- Change in environmental circumstances during the lifetime of the building e.g. if rural became industrial.

The following guidelines apply:

a. Just a gentle clean with a soft brush and mild detergent, followed by a fresh water rinse, will maintain the long-term performance of the powder coated or anodised aluminium. In rural or normal urban environments cleaning should occur every six months. In areas of high pollution, such as industrial areas, geothermal areas or coastal environments, cleaning should occur every three months. In particularly hazardous locations, such as beachfronts, severe marine environments or areas of high industrial pollution, cleaning should be increased to monthly.
b. Sheltered areas can be at more risk of coating degradation than exposed areas. This is because wind-blown salt and other pollutants may adhere to the surface. These areas should be inspected and cleaned if necessary on a more regular basis.
c. Adequate on site protection of delivered and/or installed hardware must be provided. Hardware may get knocked, scratched, or splattered with mortar, plaster, or paint during the later stages of construction. If splashes occur immediately wash down the hardware unit affected with water or methylated spirits* (*wash area thoroughly afterwards). Do not allow splashes to harden.
d. To restore powder coated surfaces that have lost gloss or are chalking, polishing with a high quality crème polish in accordance with the manufacturer's instructions is recommended. Avoid polishes that contain cutting compounds, unless the surface is extremely weathered.

DO NOT USE SOLVENTS Strong solvent type cleaners should not be used. These are harmful to the extended life of your hardware system.

CARE AND MAINTENANCE OF ANODISING

Anodised hardware is not only attractive, but also offers a durable and tough wearing finish. Some deterioration of the anodic oxide coating may occur, mainly as a result of grime deposition and subsequent attack by moisture, particularly if the moisture is contaminated with sulphur compounds.
Regular cleaning is essential to preserve the finish of anodised aluminium over a long period. The following guidelines apply:

e. Anodised aluminium should be washed with warm water and a suitable wetting agent or mild soap solution, in a similar manner to washing a car. A fine brush may be used to loosen dirt or grime. The use of anything stiffer or more abrasive may result in damage to the surface. Acid or alkali cleaners should not be used, as these will damage anodic films and may discolour coloured hardware.
f. Where greasy deposits or hard to remove grime is present, the anodising may be cleaned with a soft cloth dipped in white spirit, turpentine, kerosene, or a mild liquid scourer, followed by wiping it with a dry rag. However, the cleaner must ensure none of these solvents come into contact with other parts of the system. All solvents must be kept from contact with the Santoprene glazing gasket materials (the “rubber” seal around the glass), as most solvents will damage them.
g. It is essential to rinse anodised aluminium thoroughly with copious applications of clean water after cleaning, particularly where crevices are present, and then dry the glass to prevent water spots.

Regularly washing anodised hardware will ensure a long lasting product. In general, the following programme is recommended:

- Rural environments: every six months.
- Urban environments: every three months.
- Industrial and marine environments: every six months, as well as a monthly cold water wash.

For additional protection, especially in harsh environments, waxing with a good quality car wax after washing will assist in lifting and maintaining the appearance of your anodised hardware.
Damage to anodised surfaces may occur during building. Painters may accidentally splash paint on newly installed hardware, marring their appearance. The cleaner must act quickly and remove such splashes with a soft cloth moistened with water. Using water based paints allows the cleaner to clean with water - using solvents may put your hardware at risk.

CARE AND MAINTENANCE OF STAINLESS STEEL

Stainless steel is used for fittings and hardware for its strength, aesthetics, and its inherent high level of corrosion resistance. 2205 & 316 Grade Stainless Steel (also known as Marine Grade stainless) hardware is predominately used for external applications such as balustrades, pool fences, canopies and spider walls, but many of the entry door patch fittings and handles use 304 Grade. The design of our frameless glass systems and installation techniques are intended to make the systems as maintenance free as possible, but there are still maintenance procedures, associated with any exposed structure, that must not be overlooked. Stainless Steel is called so as it is less prone to staining - it is important to recognize that the material is not impervious to mild staining or even corrosion in some instances. The likelihood and severity of staining is a product of exposure to marine salts and other corrosive materials which can affect installations even 20km inbound from coastal areas. Pollen and other airborne matter can also contribute to corrosion so there no areas of New Zealand where the topic can be ignored. The smoother the surface is, the less prone to discoloration.
Stainless steel may be discoloured from corrosion;

- If used in areas where rain does not wash the fittings.
- If it is exposed to a more aggressive environment than that for which the particular grade of steel is intended, e.g. highly polluted air, salt solutions or residues of cleaning agents containing chlorine.
- If it has a rough surface that enables a corrosive substance to adhere to.
- If the fittings are not cleaned after installation and have oil/acid from hands.
- If the design of the steel component has crevices and narrow gaps.
- If the surface is contaminated or damaged by grinding swarf or other iron particles from tools used in the installation work.
- If fasteners of ordinary steel or dissimilar metals are used for securing the hardware, or if the hardware comes into direct contact with adjacent components made of plain carbon steel in wet or humid conditions (galvanic corrosion).

Light corrosion is often referred to as “Tea Staining” as it is a brown surface discolouration which, although not affecting the material structurally, is none the less unsightly. The likelihood of this occurring can be greatly reduced through the use of 2205 or 316 Grade, regular cleaning and the use of protective coatings.

The following guidelines apply;

a. Any stainless steel hardware should be cleaned after installation and before any glass is installed.
b. Basic cleaning can be carried out using simple soap solutions or mild detergents applied with warm water (or proprietary cleaners) and a clean non-abrasive cloth. The solution should be thoroughly rinsed off with cold water, and wiped dry with a clean absorbent cloth. Isopropyl alcohol can also be used to clean finger and hand marks.
c. Avoid bleach, as this can mark the metal surface, and avoid any abrasive applicators - especially a ferrous based cleaning pad, such as steel wool, as this can introduce contamination which reacts with the stainless steel and makes corrosion worse.
d. Protective coatings are also available that can be easily applied to the hardware which can increase maintenance periods. Information regarding these products is available from OPUS. Stainless steel hardware should be cleaned again during the final clean and if possible after any surrounding building work is complete.
e. If mild tea staining has already occurred, a plastic abrasive pad - “Scotchbrite” for example - can be used, again with warm water and a mild detergent / soap solution. When abrading however, it is important to only rub in one direction which is the same direction as any visible brushed finish. Take care to only rub the steel components, not the glass. A stainless steel rejuvenating paste can also be used and works well in combination with a scotchbrite pad. Information regarding these products is available from OPUS.
f. When installation is complete the frequency of a regular cleaning regime will vary according to the installation design and level of exposure, especially in regard to proximity to the sea. In general cleaning should take place 3 to 4 times a year. Protective coatings can prolong the maintenance interval.

In summary:

- Cleaning should be thorough and at a frequency to suit the environment
- Wash with warm water and mild detergent or soap solution or use proprietary stainless steel cleaners.
- Rinse thoroughly with clean cold water and dry.
- Do not use harsh abrasive cleaners and especially no wire wool or similar ferrous scourers.
- If mild corrosion is present, then a mild abrasive detergent or rejuvenating paste can be used with a warm mild solution and fully rinsed with clean cold water.
- Heavier levels of staining can be removed with a light plastic abrading pad (such as Scotchbrite) rubbing must only be in one direction with that being as per any visible surface finish. This is best done with a rejuvenating paste Use protective coatings after cleaning

Be careful to ensure cleaners do not effect fibre or other fitting gasket materials

CARE AND MAINTENANCE OF HANDLES / MECHANISMS

Periodic maintenance of handles / mechanisms is essential, this includes locks, handles, hinges, levers, rollers, bolts, etc. This applies particularly to mechanisms with moving parts that require lubrication. Before doing any cleaning or maintenance, you must establish exactly what hardware has been installed and how it has been constructed.

Lubrication for Mechanisms;

Mechanisms include hinges, cylinders, locks, rollers and fasteners. You can keep these mechanisms in good working order through regular cleaning and lubrication.

- OPUS recommend a Teflon-based lubricate or please refer to the manufacturer's care instructions. A soft bristle brush can be used on exposed parts. Apply Teflon-based lubricate to the moving parts - you don't need to use very much. This will limit corrosion of the exposed metals. OPUS advise to do this once every two months. If, however your hardware is near the sea or exposed to salt air, we recommend you lubricate the components once every month.

Electrical Entrance Systems;

- Do not let the product get wet.
- Keep clear of debris and general dirt.
- Wipe the keypads or swipe the device with a clean damp cloth.
- Do NOT use solvents.
- Replacement batteries will be required.

SCRATCHES AND METAL SCRAPERS

Scratches can occur from hard pointed objects or poor handling, but most often occurs from the careless removal of foreign matter from the glass surface. Mortar splatter and paint are common offenders and efforts to remove after hardening almost always lead to surface damage.

- It is essential that the foreign materials are removed before they harden. Better still, if construction work continues after glazing, that the glazed areas are protected by adhesive plastic films or suitable covers.
- One of the common mistakes made by non-glass trades people, including glass cleaning contractors, is the use of razor blades or other metal scrapers on a large portion of the glass surface. Using large blades to scrape a window clean carries considerable risk of causing damage to the glass.
- The glass industry, fabricators, distributors and installers neither condones nor recommends any scraping of glass surfaces with metal blades or knives. Such scraping usually permanently damages or scratches the glass surfaces.
- When paint or other construction materials cannot be removed with normal cleaning procedures, a new 25mm razor blade may have to be used. The razor blade should be used on small spots only. Cleaning should be done in one direction only. Never scrape in a back and forth motion as this could trap particles under the blade that could scratch the glass. Blades or scrapers can dislodge “pickup” on toughened and heat strengthened glass. There are fine particles of glass that are fused on to the surface during toughening. Once dislodged they can scratch the glass.

ONSITE CONSTRUCTION / INSTALLATION PROTECTION

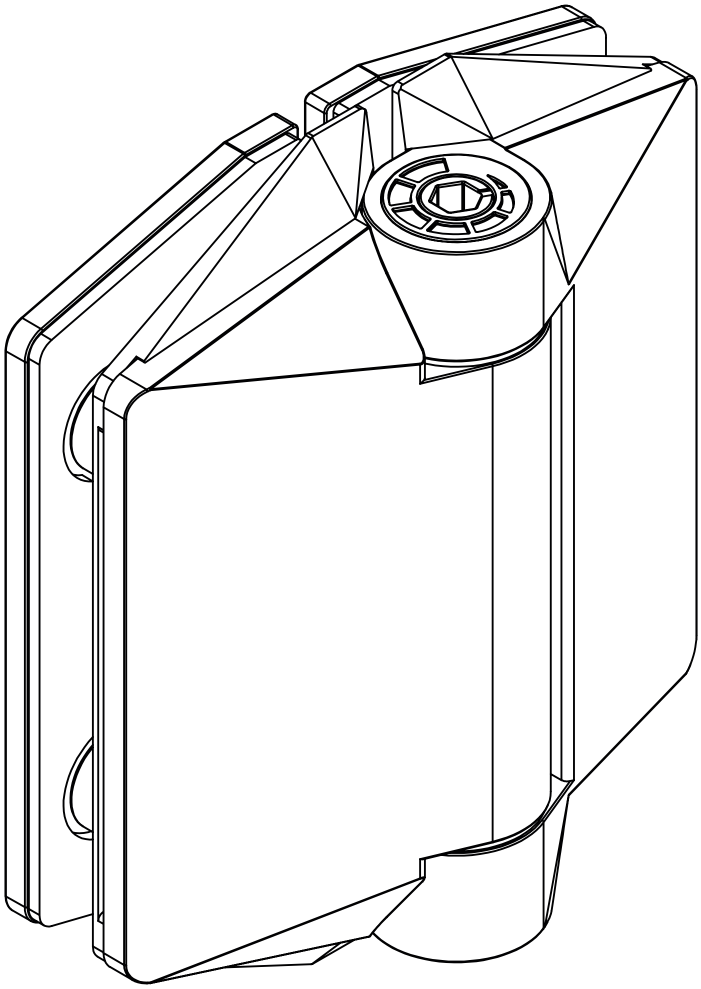
Hardware Protection during Installation:

- All the activity on a construction site means that your hardware items may get knocked or scratched, splattered with mortar, plaster, textured coating or paint during the later stages of construction.
- Please ensure that all hardware articles are masked or covered at this time. It is far easier to prevent accidents than to try and correct them. Should your hardware receive mortar or paint splashes see that these are removed before cure and follow the instructions outlined above.

For further information, please refer to the PS1 or OPUS instructions and guidelines on our website.

125 SERIES POOL HINGE

GLASS TO GLASS



FINISHES AND FUNCTIONS

- 2205 GRADE STAINLESS STEEL
- 3 SURFACE FINISHES
- SIMPLE *QUIK-ADJUST* RATCHET SYSTEM
- SINGLE ACTION
- SOFT CLOSE
- NON-HOLD OPEN
- ANTI-FOOT-HOLD GASKET AND SAFETY CAP INCLUDED

10mm
12mm

MAX.
900mm WIDE x
1500mm HIGH

MAX
MAX 45kg

— 10˚C
+ 50˚C

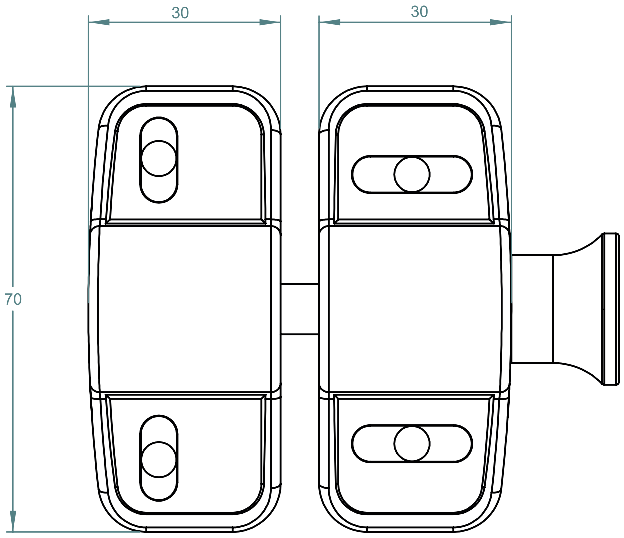
90˚

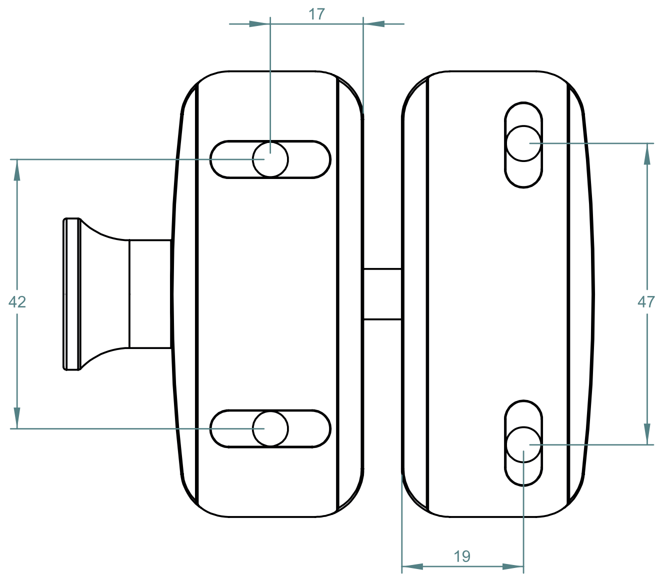
125 SERIES POOL HINGE

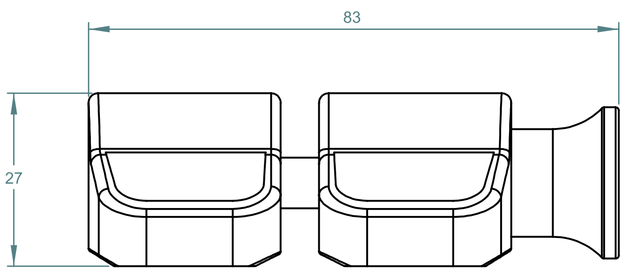
- SELF-CLOSING SAFETY HINGE FOR GLASS POOL FENCING
- TO BE USED WITH CHILD PROOF SAFETY LATCH
- COMPLIES WITH AUSTRALIAN STANDARDS FOR POOL GATE HARDWARE
- NATA ACCREDITED AND INDEPENDENTLY TESTED TO 25 000 CYCLES (NATA = NATIONAL ASSOCIATION OF TESTING AUTHORITIES AUSTRALIA)

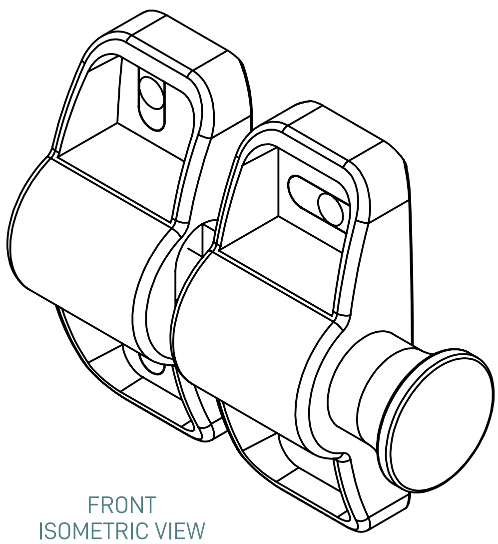
SIDE PULL LATCHES

POLYMER SIDE PULL LATCH









FRONT
ISOMETRIC VIEW